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Value Proposition of Autonomous Electric Trucks in the Freight Industry

A Qualitative Study of Transport Buyers

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Sammanfattning



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Värdeerbjudande för autonoma elektriska lastbilar inom transportindustrin: en kvalitativ studie av transportköpare

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Denna studie undersöker värdeerbjudandet för en Autonom Batteridriven Elektrisk Lastbil (ABET) inom transportindustrin, och erbjuder insikter i kunders uppfattning samt potentiella utmaningar för denna framväxande teknologi. Målet med studien är att förstå kundbehov, addressera problem relaterat till implementation, och utvärdera möjligheter för en leverantör av logistiktjänster (LSP) att nyttja ABETs. En utförlig litteraturstudie genomfördes följt av semistrukturerade intervjuer med transportköpare och experter inom industrin. Ett förslag på ABETs värdeerbjudande skapades sedan baserat på studiens resultat.

Resultaten visar att det finns en likgiltighet gentemot självkörande lasbilslogistik hos transportköpare. De ser dock dess potential som en konkurrensfördel i framtiden. Med ABETs förväntar sig transportköpare fördelar så som förbättrad transparens på marknaden, ökad operativ effektivitet och potentiella kostnadsbesparingar. För att fullt ut realisera

dessa fördelar, måste dock utmaningar relaterade till digital integration och kundernas förväntningar addresseras. Således föreslås att LSPer fokuserar på att möta grundläggande transportbehov samtidigt som fördelarna med autonoma förmågor utnyttjas.

Tillämpningar av ABETs kommer med största sannolikhet vara mest effektiva i högfrekventa flöden på korta avstånd, vanligtvis nära intermodala knutpunkter eller industriparker, vilka kommer tjäna som testfall för att uppnå en bredare användning. Aktivt deltagande i EU-piloter och forskningsprojekt kan hjälpa LSPer att bättre förstå de operativa kraven och påverka framtida standarder. I takt med att logistikbranschen utvecklas måste LSPer förbli anpassningsbara och proaktiva för att dra nytta av de möjligheter som autonom logistik erbjuder. Resultaten från denna studie presenterar ett underlag för att navigera i dessa förändringar, vilket visar att korrekt hanterad implementering av autonoma funktioner lovar betydande fördelar för tidiga användare och kan revolutionera logistikbranschen.

Nyckelord: *Autonom elektrisk lastbilslogistik, kundcentrerat värdeerbjudande, digitalisering, transportköpares behov och utmaningar*

Abstract



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Value Proposition of Autonomous Electric Trucks in the Freight Industry: A Qualitative Study of Transport Buyers

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This study explores the value proposition of Autonomous Battery Electric Trucks (ABETs) in the transportation industry, and offers insights into customer perceptions and potential challenges regarding this emergent technology. The aim of the study is to understand customer needs, address implementation issues, and evaluate opportunities for Logistics Service Providers (LSP) to adopt ABETs. An extensive literature review was conducted followed by semi-structured interviews with transport buyers (shippers) and industry experts. Based on the study's findings, a value proposition was created.

The findings reveal an indifference towards autonomous truck logistics among shippers. However, they foresee its potential as a competitive advantage in the future. Shippers anticipate benefits from ABETs such as improved transparency, increased operational efficiency, and potential cost savings.

Nonetheless, to fully realise these benefits, challenges related to digital integration and customer expectations need to be addressed. Thus, it is suggested that LSPs concentrate on meeting basic customer needs while leveraging the benefits of autonomous capabilities.

Potential applications of ABETs might be most effective in high-frequency, short-distance flows, typically near intermodal hubs or industrial parks, serving as test cases for wider adoption. Active participation in EU pilots and research projects can help LSPs better understand the operational requirements and influence future standards. As the logistics industry evolves, LSPs must remain adaptive and proactive to capitalise on the opportunities provided by autonomous logistics. The findings from this study present a roadmap for navigating these changes, demonstrating that properly managed implementation of autonomous capabilities promises significant advantages for early adopters and can revolutionise the logistics industry.

Keywords: *Autonomous Electric Truck logistics, Customer-centric Value Proposition, Digitisation, Transport buyers' needs and pain points*

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We extend our sincere appreciation to all the individuals involved in the pre-study phase, whose valuable insights and contributions enriched our understanding of the logistics sector. We would also like to extend a special thank you to all the participants who generously agreed to be interviewed for this study. Their time, willingness to share their experiences and perspectives, and candid responses were instrumental in shaping the outcomes of our research.

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Acronyms

ABET	Autonomous Battery Electric Truck
BET	Battery Electric Trucks
FMCG	Fast Moving Consumer Goods
ICE	Internal Combustion Engine
LSP	Logistics Service Providers
OEM	Original Equipment Manufacturers

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Definitions

This section provides clear and concise definitions for key terms that are used throughout this report. These definitions will serve as a point of reference that can be consulted whenever needed, to help readers better understand the concepts and ideas presented.

ABET: Autonomous Battery Electric Trucks are capable of driving themselves without human intervention, and are powered by an electric battery rather than a traditional combustion engine. These vehicles are equipped with a range of advanced technologies, including sensors, cameras, and artificial intelligence (AI) systems, which allow them to perceive their environment, interpret data, and make decisions about how to navigate and operate on the road.

Logistics service provider: Logistics service providers are defined as companies that receive and ship goods on behalf of their customers (Fritschy and Spinler, 2019).

Shipper: A shipper is a person or company that arranges the transportation of goods or merchandise from one place to another ("Cambridge University Dictionary", 2023). In the context of logistics and transportation, the term refers to the entity that is responsible for the origin of the shipment and initiates the process of moving goods from one location to another. The shipper is typically the owner of the goods and is responsible for preparing the goods for transportation, including packaging and labelling.

Transport acquirer: Transport acquirers, or transport buyers, are actors who procure transportation services from external sources. This includes shippers, freight forwarders or other entities seeking to secure transportation solutions from third-party vendors. In this report, the transport acquirer will also be referred to as the customer.

Chapter 1

Introduction

The following chapter provides an introduction to the study, aiming to provide context and a comprehensive understanding of the research topic as well as the study's limitations.

1.1 Background

Logistics is a broad field that involves managing the movement of goods, information and people (Hashimy and Rosines, 2021). The industry encompasses a wide range of activities, including planning, control, optimisation and execution of these flows, ensuring that goods are moved from their point of origin to their final destination. Logistics is a critical component of many industries, such as manufacturing, retail and Fast Moving Consumer Goods (FMCG), and the industry has become increasingly important in recent years as globalisation and e-commerce have driven demand for efficient and reliable supply chains. The rapid growth of e-commerce has revolutionised the way people shop, with shorter delivery times being the new norm. However, this trend has brought with it the need to overcome limitations on driving hours and capital usage in the trucking sector. High competition, low vehicle utilisation rates, regulations, rising fuel prices and a growing driver shortage are all hindering the sector from meeting the increasing demand brought on by the e-commerce boom. In order to stay competitive and satisfy customers' needs, the trucking industry needs to find innovative solutions to these challenges. Furthermore, the increase of traffic has a significant impact on the environment, where freight transportation is a key contributor to this issue (Hashimy and Rosines, 2021).

As a result, the logistics industry has undergone significant transformation with the development of new technologies, such as autonomous vehicles, playing a major role in shaping the future of logistics (DHL, 2023).

The use of Autonomous Battery Electric Truck (ABET) offer the potential to revolutionise the way goods are transported, leading to increased efficiency, sustainability and cost savings (Anderson et al., 2016; Simpson et al., 2019). Compared to automation, the electrification of heavy trucks has come further in both technical development and adoption as electric trucks are already in use for short-haul and urban delivery, while fully autonomous trucks are still in the testing phase, and face regulatory and technical challenges (Balasaygun, 2023; Hashimy and Rosines, 2021). However, even if the use of electric vehicles is rapidly growing, there are issues such as investment cost, battery life, technological obsolescence and charging infrastructure that need to be solved (Monios and Bergqvist, 2020). In addition, there are multiple obstacles to automation, such as safety, infrastructure, cost, user acceptance, and technical- and legal issues (Hashimy and Rosines, 2021). Despite this, remaining competitive may prove challenging for companies that do not adopt automation hence it is crucial to assess the impact of ABETs on the logistics industry and determine its potential to bring value to customers (Engström et al., 2019; Hashimy and Rosines, 2021; Sindi and Woodman, 2021). Thus, it is important to look at both automation and electrification to assess the challenges related to the implementation of ABETs.

1.2 Purpose

The aim of this study is to investigate the introduction of Autonomous Battery Electric Trucks in the transportation industry. Specifically, the purpose of this research is to delve into the value proposition that ABETs may bring and gain valuable insights into customer perceptions regarding this emerging technology.

Many people believe it will be a long time before autonomous trucks are allowed on public roads and researchers may, for this reason, not be prioritising studying this topic, or they may be focusing on other areas that are considered more pressing or closer to implementation (Monios and Bergqvist, 2020). The topic of ABETs have been the subject of extensive research over the years and even though some have touched upon the topic

of customer impact, like Osburg et al. (2022), Monios and Bergqvist (2020) and Flodén et al. (2017), there is a significant gap from this point of view. By gaining insights into what customers require and anticipate, businesses can adopt a customer-centric approach, tailor their offerings to address specific challenges, and provide enhanced value. Thus, this study aims to investigate how an ABET can meet the needs, expectations, and pain points of customers to provide significant value in their operations.

To further explore the strategies related to implementation, understanding the frictions and challenges associated with this is vital for successful deployment. By identifying and addressing potential obstacles in advance, businesses can plan effectively and allocate resources accordingly. This leads to smoother adoption and integration of the product, resulting in higher customer satisfaction and optimised return on investment.

Lastly, understanding the prerequisites required by a logistics service provider to effectively operate an autonomous vehicle solution will provide valuable guidance for businesses seeking to enter the logistics sector with autonomous solutions.

1.3 Scope & Delimitations

This study focuses on exploring the introduction of autonomous electric transport on a microeconomic level. Automated driving technologies are expected to be deployed more rapidly in trucks than in passenger vehicles due to the presence of strong use cases and business models that provide substantial economic benefits (Bösch et al., 2018; Engström et al., 2019; Sen et al., 2019). Considering this trend, the focus of this study is cantered on the logistics sector and heavy-duty vehicles. The transition towards sustainable energy sources is a pressing issue in the world today, but many countries still lack an effective and applicable environmental policy to reduce their demand for fossil fuels. In addition, some countries face adverse effects due to their weak economic situation, making it challenging for them to cope with these issues (Solaymani, 2019). While ABETs present an opportunity to reduce carbon emissions and improve efficiency, the cost of operating them compared to local labour costs will likely limit their economic viability in low-wage countries (Sindi and Woodman, 2021). Thus, the initial implementation of autonomous trucks is likely to happen in areas that have the economic stability and infrastructure

to support them, such as Europe and the United States. Thus, the scope includes an investigation of the European industry's perception of ABETs. Furthermore, although electrification is still a part of this study, the primarily focus is on on issues concerning automation.

Chapter 2

Pre-study

18 unstructured interviews was conducted to gain insight into the problem from the perspective of a Logistics Service Providers (LSP), which revealed several key insights. Read more about these interviews in *4.1 Pre-study interviews*. The interviews uncovered several key questions that remained unclear, such as the specific problems that ABETs can solve for customers, how they fit into the value chain, what frictions customers are willing to accept, and what customers expect from the current solution.

Firstly, customer acceptance of ABETs is highly dependent on the role of the customer within the organisation, such as CEO, innovation manager, transport planner, etc. Consequently, different individuals will yield diverse and potentially contrasting viewpoints of what potential value ABET can offer. It underscores the importance of engaging multiple stakeholders and considering a range of perspectives to gain a comprehensive understanding of customer acceptance and decision-making regarding autonomous truck implementation.

Secondly, while cost and reliability are important factors to customers, there is a lack of clarity about their true meaning. When it comes to cost, customers may have differing interpretations, ranging from the perception of the cheapest option to the concept of affordability in terms of value for money. For example, some customers might prioritise finding the lowest price, while others might focus on maximising services received relative to the cost incurred. Similarly, the notion of reliability can vary, where it may solely indicate a transportation service that have few disruptions or it may entail consistent and punctual delivery, ensuring goods arrive on time.

Thirdly, customers may form opinions or expectations about a technology that do not accurately reflect its current state of development or readiness. In other words, customers' perceptions of a technology may not align with its actual level of advancement, functionality or practical implementation. This highlights the importance of managing customer expectations and providing accurate information about the technology's capabilities and limitations to ensure that customers have a realistic understanding of the technology.

Complementing this, to understand the day-to-day activities of a transport acquirers' operations, observational studies was undertaken. This revealed several insights that played a crucial role in establishing the scope and delimitations of the study. Firstly, it was observed that the driver assumes responsibility for the loading and unloading processes, see Figure 2.1. While this may not be applicable to all companies, it exemplifies the expanded scope of responsibilities encompassing the driver's role, extending beyond simply operating the vehicle.



Figure 2.1: A truck driver, loading goods from warehouse to truck.

Additionally, there was a prevalent tendency to stack pallets excessively high, potentially compromising the stability of the goods, see Figure 2.2. This observation highlights a significant concern regarding adherence to rules, as it was observed that the driver often needed to re-stack pallets either to fit them inside the truck or to ensure their stability. This implication becomes particularly significant if the driver's role were to be replaced by automation. Moreover, even within the same company, various types of docking mechanisms were observed for loading and unloading purposes, indicating a lack of standardisation in this aspect.



Figure 2.2: A pallet that is stacked inappropriately with goods exceeding safe height limits, resulting in an unstable arrangement of goods.

The observations also shed light on the complex and dynamic nature of the delivery operations, which operate within ever-changing environments. For example, instances were observed where cars were parked in areas not designated for parking, or temporary containers were placed on site during ongoing construction work. Moreover, implementing changes on-site that impact the day-to-day activities of the personnel was found to be challenging.

Chapter 3

Literature study

In this chapter, a state of the art is presented based on a literature study of 80 articles and reports. This provides a brief overview of the background and context of the research, including the history of the topic. It summarises the main findings and highlights the key trends and gaps in knowledge to provide an overview of the current state of research in this field.

3.1 The Components & Flows of the Transportation System

The transportation process involves several key aspects that work together to facilitate movement of goods, products, people or materials from one location to another (Monios and Bergqvist, 2020). These considerations involve what type of item is being transported, transportation mode, and physical infrastructure such as roads, railways, ports, and airports. Systems and technologies are then implemented to manage and control the transportation process, which includes tracking and monitoring systems, scheduling software, and communication systems. Lastly, seamless operations rely on communication infrastructure, which includes internet and phone networks.

The transportation flow can be divided into three distinct parts (Hashimy and Rosines, 2021), see Figure 3.1. The first part, known as the first-mile, involves the initial stage of a shipment's journey. It begins from the point of origin, such as a factory, warehouse or distribution centre, and ends at a transportation hub, like a port, rail yard or airport, where the cargo is transferred to a

long-haul carrier. The middle-mile stage is the transportation of goods from the hub to the final destination, and lastly, the last-mile is the final segment of the shipment journey from the hub or warehouse to the customer's doorstep.

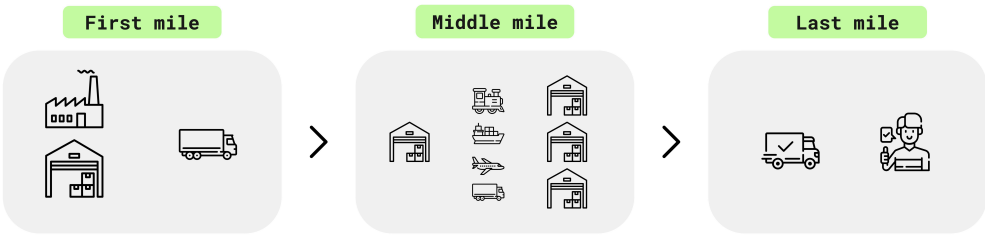


Figure 3.1: First-, middle- and last mile delivery.

Road transportation accounts for 73% of all inland freight movements within the European Union, making it a vital part of the global economy as it enables the movement of goods and people that are essential for trade, commerce and social interaction (Neuweiler et al., 2017). However, this extensive use results in frequent traffic congestion, which decreases efficiency and raises overall expenses. The transportation industry today is highly competitive, and the service of transporting goods from one location to another has become more of a standard commodity that is offered on an open market (Abrahamsson et al., 2023). Consumers have become more demanding and cost-conscious, and transportation services are increasingly being evaluated based on price and delivery time. As a result, transportation providers are facing high pressure as they compete for customers in an open market. The challenge of cost is directly linked to the chosen mode of transportation (Neuweiler et al., 2017), however Flodén et al. (2017) found that rail is avoided even if other conditions are equal. One possible conclusion that can be drawn from this is that this might partially explain the high utilisation of road transportation.

3.2 Transportation acquisition

LSPs are experiencing an increase in demand for faster deliveries and as a result, they are under pressure to deliver products within a specified time frame (Sindi and Woodman, 2021). Failure to deliver within the agreed duration can result in penalties, which could have financial implications (Engström et al., 2019). Thus, transport buyers consider many factors when choosing a

transportation service in terms of quality. However, transport quality is a wide factor that could include many things such as time, reliability, frequency, risk of damage, etc., which can be summarised in that it is critical to ensure goods and products are transported from one place to another efficiently and reliably. According to Flodén et al. (2017), cost, transport quality, transport time and reliability are the most important factors that a transport buyer would consider when choosing a transportation service, where cost is the number one factor that would influence the decision-making process. This is not surprising given that most aspects of operations are related to cost, such as the driver, accounting for 49.3%, and fuel, 17.7% (Sindi and Woodman, 2021). Across Europe and over the past 30 years, the main factors that influence transport acquisition have remained the same, despite the changes in the transportation industry and the advancement of technology. An observation that Flodén et al. (2017) makes is that the importance of reliability seems to be reduced for longer transport times because transport buyers would prioritise cost over reliability when it comes to longer transportation routes.

3.3 Navigating Industry Challenges: Drivers shortage

The transportation industry is currently facing a significant challenge in terms of a shortage of drivers, which has been increasing over the last decade (Chen and Lu, 2021; Sindi and Woodman, 2021). This problem has been worsened by the rise of e-commerce and the need for a more flexible scheduling to cater for same-day delivery and reduced turn-around time. Moreover, truck driving is becoming less attractive as a job, particularly among younger generations (Neuweiler et al., 2017). The shortage is partly due to certain EU countries maintaining a minimum age requirement of 21 for truck drivers, while most bus and coach driver roles across the EU exclude young individuals seeking their first job by setting an age requirement between 21 and 24 (IRU, 2022). On top of this, the industry is facing wage concerns, poor working conditions and health issues which, when combined with high license and training costs, does not favour the profession (Hashimy and Rosines, 2021). With an ageing population, there is a growing gap between retirement and new drivers, which is said to triple the rate of unfilled truck driver positions (IRU, 2022). This means that Europe could lack over two million drivers by 2026 if unattractive factors, such as demanding hours for long periods of

time away from home, are not addressed (IRU, 2022; Sindi and Woodman, 2021). It has been suggested that ABETs could solve this challenge as trucks equipped with automated driving technologies can travel the majority of the journey without the intervention of a driver (Engström et al., 2019; Sindi and Woodman, 2021). Over the last 10 years, there has been a significant increase in automated technology in road transportation, for example cruise control and lane assist, as digitisation have made it possible to operate vehicles without human intervention (Neuweiler et al., 2017; Simpson et al., 2019). However, several barriers has to be overcome before fully autonomous trucks are commercially available.

3.4 Electrification of the Transport Industry

In 2016, the World Health Organization stated that air pollution is the most pressing environmental hazard towards human health. As a result, there has been a demand to reduce CO2 emissions across all industries, which has become one of the biggest challenges the world is facing (Solaymani, 2019). The need for sustainability has thus resulted in significant changes in production and consumption patterns, with businesses facing demands and constraints to ensure their operations align with sustainability goals (Dyck and Silvestre, 2018; Hashimy and Rosines, 2021; Kneipp et al., 2019).

Trucks and buses are the biggest emitters on Europe's roads (European Commission, 2023), contributing to approximately 27.3% of the total greenhouse gas emissions in the EU (Hashimy and Rosines, 2021). In 1999, van Veen-Groot and Nijkamp identified a trend towards using less environmentally friendly modes of transport. They warned that if this trend were to continue, it could result in unsustainable outcomes for the environment and emphasised the need for action to address this issue. Today, there is extensive research and development towards electrification as a means to tackle the issue, but despite these efforts, the unsustainable trend remains as there still is a lack of demand for fossil-free transportation (Abrahamsson et al., 2023). Even though some customers may prefer or value sustainability, there are not enough of them to create a large market for fossil-free freight. As a result, logistics companies may not see a financial incentive to provide such services, leading to a gap between the demand for and supply of sustainable transportation. The European Union has taken steps to address this issue, including the introduction of legislation that mandates the reduction of CO2 emissions by 45% in 2030, 65% in 2035 and 90% in

2040 from heavy-duty vehicles, demanding high shares of zero-emission vehicles (European Commission, 2023).

One way to decrease the environmental impact of logistics and transportation is by utilising an eco-friendly fleet of vehicles since conventional diesel-powered trucks are known to contribute significantly to greenhouse gas emissions and air pollution (Hashimy and Rosines, 2021; Sen et al., 2019; Solaymani, 2019). Factors such as fuel economy, payload and vehicle-miles-travelled play a crucial role in determining energy consumption and emissions, where Battery Electric Trucks (BET) have emerged as a promising alternative. Studies have shown that BETs are more energy-efficient and emit less greenhouse gases than both diesel-powered- and natural gas-powered trucks (Sen et al., 2019). However, this positive effect is partially offset by the indirect health impacts associated with power generation from electricity sources that can have adverse effects on human health. If the electricity used to charge the batteries comes from an energy mix that relies heavily on for instance coal or natural gas, then the carbon emissions reduction benefits of electric vehicles powered by batteries may be limited or even non-existent.

A major drawback of BETs is their cost. Electric heavy-duty trucks can cost 2-3 times more than their diesel counterparts, and with fluctuating electricity prices, the total cost of ownership is hard to determine, making them a more expensive investment for businesses (Abrahamsson et al., 2023; Roland Berger, 2015). Despite this, the potential long-term cost savings and environmental benefits make them an attractive option for companies looking to reduce their carbon footprint and operating costs. For example, the quiet nature of BETs makes them suitable for use in areas with noise regulations. This can potentially increase their utilisation rate as they can operate during nighttime hours without disturbing the surrounding environment (Abrahamsson et al., 2023).

One of the challenges facing BETs is how to navigate long haul shipping due to charging (Balasaygun, 2023). Plug-in charging has come under increasing criticism due to the 1-2 hour charging times required (Abrahamsson et al., 2023). A potential solution to this problem is battery swapping, where depleted batteries are replaced with fully charged ones at specialised charging stations, which can take as little as 3-5 minutes. However, the implementation of battery swapping requires a network of stations with multiple charged

batteries, which represents a major infrastructure challenge.

Although not all autonomous vehicles will be electric in the immediate future, the two technologies work together and can complement each other (Monios and Bergqvist, 2020). The success of electric and autonomous vehicles will depend on how well they fit into existing logistics systems, rather than trying to force these systems to change to accommodate the technology (Abrahamsson et al., 2023). This new system will lead to new relations between actors, thus the business model of ABETs need to be considered, looking into all dimensions of sustainable innovation (Monios and Bergqvist, 2020).

3.5 Transforming Logistics: Embracing a Holistic View

The commercial freight industry have the benefit of fleet size, the potential to better utilise freight transport lead time, and the potential for greater return on investment compared to passenger vehicles. This makes the adoption of autonomous technology more financially viable for commercial freight companies, which may lead to them being early adopters of the technology (Chen and Lu, 2021; Sindi and Woodman, 2021). Thus, truck manufacturers are starting to consider autonomous vehicles as a practical option instead of traditional manual trucks. However, this requires a systems view of the entire supply chain to ensure that sustainability is integrated throughout the process (Szekely and Strebel, 2013). The current electrification and automation of the road freight industry can be viewed as two simultaneous transitions from one socio-technical system to another (Forsselius and Ottosson, 2022). This shift involves modifying multiple components that make up the socio-technical system, including not only the trucks themselves, but also the supporting road and fuel infrastructure, regulations and user practices. Therefore, the transition encompasses more than just the replacement of trucks.

3.6 The Evolution of Automation in Logistics: From Indoor Systems to Autonomous Trucks

Indoor connected and automated systems have been used in logistics for several decades, starting with in-house logistics applications in the 1950s (Flämig, 2016; Hashimy and Rosines, 2021). The development of automation in logistics has since been more prevalent and while autonomous vehicles have been in use in controlled environments for years, major trucking companies only began making significant efforts to bring autonomous trucks to the open road around 2010 (Chen and Lu, 2021). In recent years, the European Commission mandated certain safety features for newly registered trucks, such as lane departure warning systems and advanced emergency braking systems (Roland Berger, 2015). Additionally, the drivers of automation are rooted in technological advancements, such as AI, machine and computer vision, geo-localisation, big data, and advancements in radars, which has led to enhanced operational performance (DHL, 2023; Hashimy and Rosines, 2021). These technologies are enabling greater precision, efficiency and safety in logistics operations, paving the way for automation in the industry.

3.7 The Roadmap to Autonomy: Understanding the Levels of Automation

The Society of Automotive Engineers (SAE) levels of automation are a framework that categorises autonomous vehicles based on the level of automation they offer (SAE International, 2021). The levels of automation are designed to help clarify the capabilities and limitations of autonomous vehicles, and to provide a common language for discussions surrounding the development and deployment of autonomous technology. There are 6 levels in total:

Level 0 - No Driving Automation: The driver is in full control of the vehicle, and there is no automation present.

Level 1 - Driver Assistance: The vehicle includes some automated features, such as adaptive cruise control or lane-keeping assistance, but the driver is

still responsible for most driving tasks.

Level 2 - Partial Driving Automation: The vehicle can control both steering and acceleration/deceleration, but the driver is still responsible for monitoring the environment and taking over when needed.

Level 3 - Conditional Driving Automation: The vehicle can perform most driving tasks, but the driver is still required to be present and ready to take over if needed in certain situations.

Level 4 - High Driving Automation: The vehicle can operate autonomously in most situations, but may still require driver intervention in certain circumstances.

Level 5 - Full Driving Automation: The vehicle can operate autonomously in all situations and does not require any human intervention.

3.8 Predictions of Autonomous Deployment

Between now and 2030, it is predicted that levels 3-5 of automation are expected to be increasingly adopted in the market (Hashimy and Rosines, 2021; Raj et al., 2020; Sindi and Woodman, 2021). However, level 4 autonomy is considered to be just an intermediate step and the actual transformation will happen when fully autonomous vehicles without drivers (level 5 autonomy) are introduced, enabling scaling up to large numbers (Monios and Bergqvist, 2020). Automation is becoming increasingly important in many industries. Some of the benefits are the removal of human error and shorter lead times, where for instance autonomous trucks have the potential to address the issue of driver shortage (Neuweiler et al., 2017; Sindi and Woodman, 2021). Yet, it is unlikely that driverless trucks will be used on public highways in the near future (Engström et al., 2019). Therefore, it is expected that low-speed automated vehicles that can navigate complex traffic situations in restricted areas will be the first to achieve high market adoption (Hashimy and Rosines, 2021). Also, the adoption of automated trucks may vary depending on the region, depending on factors such as regulatory compliance, infrastructure, and the research and development efforts of other industry players, such as Original Equipment Manufacturers (OEM). With multiple large vehicle manufacturers and suppliers of vehicle components, as well as an early

emergence of testing and regulatory landscape, Europe is expected to be at the leader of Connected Automated Vehicle adoption. Nevertheless, independent of how and where autonomous technology will emerge, it can be asserted that the trucking and logistics industry will undergo a significant transformation in the upcoming years, presenting both risks and opportunities for the parties involved.

3.8.1 The Role of Intelligent Software in Achieving High-Level Autonomy

The transportation system are most effectively represented by a network structure, where information technology and intelligent software are at the centre of the transport market (Monios and Bergqvist, 2020). To achieve high-level autonomy (SAE 4-5), an autonomous vehicle must possess the capability to gather information about its surroundings, process the data, adapt to changes in the operating environment, avoid errors, optimise routes, and automate manual operations for successful operation (Hashimy and Rosines, 2021; Osburg et al., 2022). This involves having a guidance control system for order assignment and route planning, as well as a communication system (Flämig, 2016). Consequently, the software used in autonomous trucks will be crucial and contribute significantly to the value of the product (Hashimy and Rosines, 2021). Companies that invest in developing software for autonomous trucks will thus be able to earn high profits in comparison to hardware, that most likely will become standardised and therefore not provide a competitive advantage.

3.9 Exploring the Advantages of Automation

Automation could potentially benefit the transportation industry in numerous ways. One of the primary gains is that it leads to a significant reduction in emissions (Fagnant and Kockelman, 2015; Hashimy and Rosines, 2021). The transport industry accounts for a significant percentage of greenhouse gas emissions, which is endangering both the environment and human health. Automation can increase road capacity, enabling more vehicles to travel on the same roads at the same time by allowing vehicles to drive at a close distance between them. As the space between the vehicles is reduced, there is a

decreased risk of traffic jams. This can help to reduce congestion and improve overall traffic flow, leading to shorter travel times, less fuel consumption and lower emissions. This can make transportation more efficient, reduce the economic cost of congestion and improve air quality by reducing the amount of time vehicles spend idling in traffic (Anderson et al., 2016; Fagnant and Kockelman, 2015; Greenblatt and Shaheen, 2015; Sen et al., 2019). With this, transportation companies will experience lower fuel costs at the same time as fleet utilisation also can be optimised.

Safety is another benefit of automation as human error is one of the leading causes of road accidents (Fagnant and Kockelman, 2015; Hashimy and Rosines, 2021; Roland Berger, 2015). Because automated trucks has to be equipped with advanced safety features and are programmed to follow traffic rules and regulations strictly, the risk of accidents caused by speeding, distracted driving and other human factors are consequently reduced. Additionally, automated trucks can detect and respond to potential dangers on the road more quickly and effectively than human drivers, further increasing safety on the road (Engström et al., 2019).

Labour costs, which include not only wages but also benefits such as health insurance, retirement plans and other expenses associated with employing people (Hashimy and Rosines, 2021; Simpson and Mishra, 2021), can fluctuate based on factors such as supply and demand for workers, minimum wage laws and the cost of benefits. Therefore, companies often look for ways to reduce labour costs while maintaining productivity, which is where automation can be a beneficial solution (Engström et al., 2019). By eliminating the need for human drivers, companies can save on these expenses on top of benefiting from increased reliability and flexibility. For example, automated trucks can travel for long periods of time without needing breaks, which allows more goods to be transported in less time.

3.10 Challenges & Opportunities in the Deployment of Automated Trucks: Early Adoption Strategies

The advantages of using automated trucks are often described in general terms, but vary significantly depending on the particular use case that is

considered (Engström et al., 2019; Flämig, 2016). For example, a likely scenario is that an autopilot would take over driving in areas of heavy traffic congestion and/or tight delivery zones that initially were designed for smaller vehicles. The truck would park or dock itself without the need for the driver to manually manoeuvre it, which could help prevent damage that otherwise might occur. However, the current general belief is that automated technology will have the most significant opportunities on long-haul freight (Engström et al., 2019). One such scenario is platooning, a term used to describe a convoy of vehicles that are closely spaced and move together on the road (Engström et al., 2019). A fully automated platooning convoy of trucks may only need a human driver in the lead truck, while the following trucks can operate unmanned (Chen and Lu, 2021). Studies have shown that platooning is a promising technology that can provide a fast payback on investment, where one of the main advantages is its potential to reduce fuel consumption and lower carbon emissions (Roland Berger, 2016; Sindi and Woodman, 2021). The cost of fuel is one of the highest operating expenses in the haulage industry and Volvo claims that if the computer-controlled trucks follow each other just one second apart, there is a 10% fuel consumption improvement (Sindi and Woodman, 2021). Several other studies have also shown a positive impact on the environment with platooning, saving up to 20-30% of fuel consumption (Neuweiler et al., 2017). In addition to fuel savings, platooning also posses the possibility of increasing traffic flow and improve safety (Sindi and Woodman, 2021). The current focus for on-road platooning are being done on an SAE level 3 automation, but the current state of the art is believed to be insufficient to support this level of automation for mixed traffic and highway-speed operations (Engström et al., 2019).

3.10.1 Regulatory Barriers

Because of the complexity of automation technology, regulatory barriers and the need for public acceptance, the initial deployment of autonomous trucks are expected in highly constrained environments with little traffic, such as ports, distribution centres and production plants (Flämig, 2016; Hashimy and Rosines, 2021). Therefore, shippers with large private fleets with consistent tractor manufacturers and predetermined routes, as well as large carriers operating long-haul trucks on fixed routes between terminals, are expected to be the main early adopters of automated trucking applications.

Regulations are necessary to protect public safety, but they can hinder technological advancement if they are overly restrictive, slow to adapt or create barriers to entry (Hashimy and Rosines, 2021). Thus, before high levels of automation (level 4 and 5) become common, many vehicles will rely on both human drivers and automation systems to get approval for public road use. However, given the potential economic benefits and public interest in automation, it is reasonable to expect rapid technological progress according to Simpson et al. (2019). Nonetheless, there is a slow development of automated heavy duty trucks that does not match the optimistic portrayals in media, which could lead to unrealistic expectations from the public and policymakers (Abrahamsson et al., 2023). This is a significant disadvantage as the benefits will not be realised as quickly as they could be, making it difficult for companies to plan and invest in automation since they may not be confident that the technology will be available as expected. Furthermore, without the presence of a detailed business model making a strong case for investment, the presentation of use cases are trivial regardless of how much they promise (Flämig, 2016). This is further strengthened by Neuweiler et al. (2017), who found that customers were interested in seeing new and innovative solutions or technologies introduced in the transportation industry, but the transportation companies were not willing to invest in developing these new ideas themselves. Instead, they only wanted to adopt and implement innovations that had already been developed and tested by others. Furthermore, using predictions and scenarios in such a complex setting means that the use cases cannot be modelled with certainty (Fritschy and Spinler, 2019). Still, there seems to be a universal consensus amongst several players that autonomy in the freight industry should be realised (Engström et al., 2019).

3.10.2 Public Acceptance

The implementation of an ABET system faces several barriers that may impede its progress when viewed from the perspective of transport buyers (Klumpp, 2018; Litman, 2022; Nastjuk et al., 2020). Emerging technologies face uncertainty related to market acceptance or rejection, especially when it comes to radical innovations that are not yet widely understood or tested (Kohl et al., 2018; Nastjuk et al., 2020). Perception can shape how the public views the potential risks and benefits associated with a technology. Therefore, trust is a significant factor influencing the acceptance of autonomous vehicles,

meaning that events and accidents involving ABETs may lead to fear and hesitation towards the technology. However, the potential reduction in accidents caused by human error may sway public opinion towards favouring autonomous trucks (Kohl et al., 2018; Litman, 2022; Nastjuk et al., 2020). Moreover, if ABETs make significant advancements and improvement of the technology, it is likely that they will earn the trust of the public and receive widespread recognition for their advantages. However, if there is a lack of progress in their technological capabilities, ABETs are more likely to attract unfavourable publicity (Simpson et al., 2019). Trust for the technology could also be built upon trust for regulatory authorities and laws that ensure safety, yet safety restrictions can be perceived as a loss of control, which may hinder public acceptance (Kohl et al., 2018). Thus, there is a need for significant investments in technology, infrastructure and education, where governments, industries and other stakeholders must collaborate to integrate ABETs in the current infrastructure (Simpson et al., 2019). For instance, regulations that limit the number of hours a driver can drive continuously may not be relevant to a driverless vehicle.

3.11 Tackling Environmental Challenges with Sustainable Innovation

Innovation is about creating something new and changing the way things are done (Szekely and Strebel, 2013), having the potential to transform behaviours, although this is a difficult task to accomplish (Shove, 2003). The success of an innovation depends on several factors, including its reinvention potential and perceived complexity (Simpson and Mishra, 2021). For example, innovations that are flexible and adaptable are more likely to be perceived as advantageous and compatible with potential adopters' needs (Simpson et al., 2019). However, the success of the technology in the market also depends on external factors beyond the company's control, such as governmental influences, policies and regulations.

The logistics sector is becoming increasingly technology-driven and companies are required to adapt to new market trends. ABET technology is still too immature to compete with the existing system, for example the highly efficient network of fuel stations that currently exists compared to charging infrastructure (Forsselius and Ottosson, 2022). Thus, even if autonomous

trucks were widely available today, it would still take at least ten years to fully replace the current logistics system (Simpson et al., 2019). This highlights the importance for a new technology to be adapted to what the customer wants in order to encourage adoption, especially for segments that are set in their ways (Simpson and Mishra, 2021; Szekely and Strebel, 2013). Ease of comprehension is key to fast adoption, and introducing innovation in smaller, incremental steps can make it easier for people to understand it. For instance, the transition to biofuel has been shown to be relatively easy as long as freight companies can use existing vehicles and infrastructure. Thus, the lack of charging stations is hindering long-haul transportation with electric trucks ((Abrahamsson et al., 2023; Nastjuk et al., 2020).

3.11.1 The Social-, Economic- & Environmental Dimensions

Achieving a successful innovation that balances all three dimensions of sustainability can be challenging because when one dimension is improved, it can lead to a decrease in performance in another (Szekely and Strebel, 2013). Therefore, it is crucial to consider the interrelationships and trade-offs between these dimensions.

The social sustainability dimension refers to the ability to meet the social needs of stakeholders, while also ensuring social justice and equity for individuals, communities and society as a whole (Szekely and Strebel, 2013). It recognises that sustainable development is not just about protecting the environment but also about creating positive social outcomes that benefit everyone. For example, ABETs have the potential to reduce accidents, noise and congestion which benefits society (Raj et al., 2020; Simpson et al., 2019).

One of the major concerns resulting from automation is job losses as advanced technology often requires higher-skilled jobs (Engström et al., 2019; Sindi and Woodman, 2021). However, the combination of automation with electrification has the potential to create new employment opportunities in areas such as construction, maintenance and operation of charging stations (Raj et al., 2020; Sen et al., 2019). Additionally, as the technology continues to develop, it is likely that ABETs will still require human support for tasks such as loading and unloading goods, and dealing with any issues that may arise while driving (Sindi and Woodman, 2021). While the driver's job may

change towards managing the delivery process and overseeing vulnerable parts of the process, the social cost of job losses and degradation must be accounted for (Flämig, 2016; Fritschy and Spinler, 2019). Freight companies will need to consider the impact the adoption of ABETs will have on their staff and potential reputation damage resulting from loss of employment opportunities (Fritschy and Spinler, 2019). However, companies can consider re-skilling their current workforce and transferring them to new roles to minimise the possible impact of job losses (Sindi and Woodman, 2021).

Economic sustainability refers to the ability of a system to generate and maintain economic growth and long term prosperity (Sen et al., 2019). In the transportation industry, the high costs associated with driver salary, accidents and fuel costs pose a significant challenge to achieving economic sustainability (Simpson et al., 2019; Sindi and Woodman, 2021). However, the development of ABETs offers a potential solution to this challenge (Fritschy and Spinler, 2019; Sindi and Woodman, 2021). It has been estimated that the adoption of automation could lead to cost savings of up to 30%, including the elimination of driver costs (Roland Berger, 2016; Sen et al., 2019). Yet, because the transportation industry typically operates on small margins, there is a significant pressure to realise quick and certain returns on investments in new technologies, which could hinder the adoption of ABETs (Engström et al., 2019).

Environmental sustainability involves developing products, processes and services that minimise negative impacts on the environment while promoting the efficient use of resources (Kneipp et al., 2019). The need for systemic change to existing production and consumption patterns is widely recognised as a crucial step towards achieving environmental sustainability (Engwall et al., 2021). Looking at autonomy through the environmental dimension, ABETs can enable the optimisation of truck operating hours, potentially leading to a decrease in the number of trucks on the roads, and subsequently resulting in positive environmental outcomes (Fritschy and Spinler, 2019). However, as a consequence of lower costs, the implementation of ABET technology may lead to a rebound effect, causing an increase in travel and freight demand. This would result in higher energy consumption rates and potential public health problems, with the increase in the number of trucks on the road (Fritschy and Spinler, 2019; Sen et al., 2019).

3.12 Business Model Innovation: the Complexity of Challenging Existing Business Models

ABETs are revolutionising the transportation industry and opening up new opportunities for businesses to explore. However, taking advantage of this technology requires developing new skills and capabilities, as well as rethinking traditional business models (Abrahamsson et al., 2023; Fritschy and Spinler, 2019; Monios and Bergqvist, 2020). It is projected that by 2040, autonomous trucks will be shared among multiple LSPs, and as a result, it is likely that only a few large players will dominate the market and that only a few platforms will ultimately cooperate directly with OEMs. Thus, it is crucial for companies to innovate their business offerings to remain competitive (Engström et al., 2019; Monios and Bergqvist, 2020).

When challenging its existing business model, an organisation must consider systematic changes that transform how they are creating-, delivering-, and capturing value (Adams et al., 2016; Engwall et al., 2021; Szekely and Strebel, 2013). Because of the transformative impact of autonomous technology, many players in the market will be affected by its implementation (Fritschy and Spinler, 2019). Stakeholders from different sectors will therefore need to collaborate in order to address concerns about the deployment and business models surrounding automation (Engström et al., 2019; Raj et al., 2020). Still, it can be hard to break into the market with a new business model as the existing system has institutionalised lock-ins (Engwall et al., 2021). Disruptive technology may face delays in adoption due to market uncertainty and resource scarcity, as well as a weak value proposition in the existing business model, particularly when it is based on older technology. In other words, businesses may struggle to adopt new technology if they have invested heavily in the old one, and the benefits of the new technology may not be clear to all stakeholders (Monios and Bergqvist, 2020). One of the major challenges in the logistics industry is that the business models are typically designed for specific and highly varied transportation operations. They require different sets of deployment factors, such as technology, infrastructure and workforce, which makes it difficult to create a standardised approach to automation deployment (Engström et al., 2019).

3.13 Navigating Adoption Challenges: Overcoming Physical, Psychological & Policy Barriers

Negative factors, like high cost or distrust, have greater impact on adoption rates than positive factors do. Therefore, it is important to review all types of obstacles for adoption, such as physical-, psychological- and policy barriers (Raj et al., 2020; Simpson et al., 2019).

The physical barriers are mostly related to autonomous vehicles' ability to navigate safely in the public space (Flämig, 2016; Hashimy and Rosines, 2021). For example, during the initial transitional period where both autonomous and conventional vehicles coexist, it may be necessary to develop various traffic network strategies, such as dedicated lanes to ensure the safe and efficient integration of autonomous vehicles into existing traffic systems (Flämig, 2016; Raj et al., 2020).

The psychological barriers are rooted in public perception of the technology. Understanding the emotional responses to autonomy is therefore an important step towards integrating ABETs into the current system as it is one of the most prominent barriers to implementation (Osburg et al., 2022; Raj et al., 2020). To help with public acceptance, tests and trials can demonstrate the evolving capabilities of the technology (Engström et al., 2019), establishing trust, viability and credibility of the technology. However, even with widely-used technologies like automatic parking and blind spot detection, scepticism persists, illustrating that it takes time to change people's opinions (Neuweiler et al., 2017).

The most significant impacts resulting from automation are likely to be seen in the way that humans interact with machines, which probably will have a significant impact on many industries and professions (Osburg et al., 2022). While machines are capable of processing vast amounts of data and performing certain tasks with speed and accuracy, humans bring a level of creativity, critical thinking, and emotional intelligence that machines cannot replicate. A psychological factor related to this is the loss of control (Osburg et al., 2022). In many cases, drivers are viewed as a crucial safety backup measure and the idea of removing them altogether can therefore be met with resistance (Fritschy and Spinler, 2019). The driver also adds value to

operations, such as detecting issues or securing the goods. Nevertheless, as autonomous technology continues to improve, the capacity to substitute or automate tasks traditionally performed by drivers is not considered a significant problem or obstacle (Flämig, 2016; Sindi and Woodman, 2021).

Lastly, the implementation ABETs has important legal and safety implications that need to be addressed, beyond the physical- and psychological barriers (Chen and Lu, 2021; Fagnant and Kockelman, 2015). Safety regulations will be a crucial factor as safety cannot be compromised for economic savings (Engström et al., 2019; Roland Berger, 2016). For instance, the Vienna Convention stipulates that the driver must maintain control over the vehicle at all times, highlighting the importance of ensuring that the technology is safe and reliable, and that regulations are not up to date with autonomous technology (Flämig, 2016). This requires extensive certification and testing to establish industry standards (Hashimy and Rosines, 2021; Raj et al., 2020). Yet, it appears that politicians and policymakers rely heavily on the market to address safety concerns (Abrahamsson et al., 2023). There are some exceptions where governments have taken action. For instance, in 2020, the British Standards Institution released a specification, PAS 1881, "Assuring Safety of Automated Vehicle Trials and Testing", to support the safe public trials and development testing of driverless vehicles (Hashimy and Rosines, 2021).

Research shows that human errors are responsible for over 90% of crashes, and a significant portion of fatal accidents involve driver distraction, fatigue or alcohol and drug use (Fagnant and Kockelman, 2015). While human drivers are typically not held at fault for circumstances beyond their control, autonomous vehicles may still face legal challenges as they have the potential to make more informed decisions. This raises questions about liability when accidents involving ABETs occur. The ethical considerations around the moral code for decision-making vary across regions and countries, further complicating the issue (Hashimy and Rosines, 2021). To promote public trust, it is therefore crucial to have liability frameworks and safety measures in place. Deciding on who will be responsible for accidents and how liability will be distributed among stakeholders is critical for the successful implementation of autonomous technologies (Flämig, 2016; Raj et al., 2020; Sindi and Woodman, 2021).

3.13.1 Required Capabilities: Addressing Adoption Challenges for Autonomous Technologies

Organisations are highly diverse in their characteristics and behaviours, which makes it difficult to predict their adoption of new technologies (Simpson et al., 2019). If autonomous technology fails to evolve quickly in the coming years, it may struggle to gain widespread acceptance. Therefore, it is crucial to focus on addressing the most significant barriers to adoption and accelerating the pace of technological advancement in order to ensure that the potential benefits of autonomous technologies can be fully realised (Raj et al., 2020; Simpson et al., 2019).

The transition to electric service offerings and software capabilities has presented significant challenges for traditional automotive market roles. This has opened the door for a range of new players to enter the industry, including software and service giants, as well as new startups (Sindi and Woodman, 2021; Tian et al., 2016; Yin et al., 2017). In the autonomous truck market, software will be the key differentiating factor, making hardware standardised and a commodity (Fritschy and Spinler, 2019), even though hardware is still important to develop as without it, the ABET will not be a truck at all. Many people in the transportation industry agree that incorporating software and data elements into their offerings is the key to providing better and more efficient transportation services in the future. Thus, system security and integrity have emerged as significant concerns (Monios and Bergqvist, 2020; Raj et al., 2020). However, there is currently a lack of criteria for verifying the safety and cyber-security of autonomous vehicles, which poses a significant barrier to successful implementation (Hashimy and Rosines, 2021).

The primary reason why trucking companies invest in automated driving is to increase profit margins (Engström et al., 2019). Automation is expected to streamline processes, which could result in companies being able to offer products or services at lower prices than competitors. Though, while cost is a significant factor in transportation acquisition, consumers are increasingly concerned with ethical issues, making products and retailers that have committed to these standards more attractive (Hashimy and Rosines, 2021). As ABETs have the potential to significantly impact the sustainability dimensions, companies that invest in automation will need to take ethical considerations into account to stay relevant and attract customers who are willing to pay a premium price for sustainable and responsible products and

services (Flodén et al., 2017; Hashimy and Rosines, 2021). Yet, focusing solely on sustainability may not necessarily provide a competitive advantage in all market segments. While customers may express concern for sustainability, it is not always a driving factor in their purchasing decisions. Thus, it is important for businesses to understand the priorities and values of their target market to determine the most effective way to differentiate themselves and remain competitive in their respective industries (Engström et al., 2019).
Once autonomy is fully developed, it is likely that electric vehicles already have a high market uptake (Sen et al., 2019). Therefore it is more beneficial to focus on developing automation rather than electrification. Smaller actors are anticipated to drive this technological advancement, whereas larger ones are likely to adopt it by modifying their business models (Monios and Bergqvist, 2020). ABETs might move slower than conventional trucks due to their safety systems, so it is necessary to implement autonomy in environments that can handle this (Sindi and Woodman, 2021). For this reason, it is most probable that autonomous vehicles will be introduced in closed transport systems, such as fenced areas, airports and ports, where platooning is the most feasible use case to start the autonomous journey (Monios and Bergqvist, 2020; Flämig, 2016; Roland Berger, 2016).

Chapter 4

Method

The project process involved five stages to explore and address the problem, see Figure 4.1. It began with a pre-study, including interviews and observations, to understand the industry and current practices. The insights from this stage led to the synthesis of a clear problem definition. Data was then collected through semi-structured interviews to understand customer requirements and preferences. In the final stage, an implementation plan and business model case were developed based on the analysis of the results.



Figure 4.1: Research design and project process.

4.1 Pre-study interviews

The pre-study served as an initial exploration and preparation phase to understand the problem domain. It gave a solid foundation for formulating the research problem by assessing the knowledge of the industry, which

facilitated the identification of the current state of the transportation market and directions for further investigation. Moreover, it provided an evaluation of the proposed study in terms of available resources, time constraints and potential challenges.

Through a collaborative effort with a logistics company, 18 LSP representatives could be reached. Unstructured interviews, based on a few discussion points, were conducted to gather information about the representatives' understanding of customer needs and to ascertain the areas of interest or specific topics that the participants sought to explore further. Each interview was about 45 minutes long, where both researchers took notes and asked questions. The notes were then compared to formulate a summary. The information obtained from the interviews were subject to an inductive thematic analysis to gain a deeper understanding of the problem (Braun and Clarke, 2006). The analytical process involved identifying patterns in the data and grouping statements from the interviewees to uncover overarching themes.

4.1.1 Observations

To complement the interviews, four non-participatory open observational studies (Liu and Maitlis, 2010) were conducted. This involved directly observing workers in their natural environment without actively participating or influencing the situation or the participants to gain a deeper understanding of their behaviours and actions. Two observations involved accompanying truck drivers throughout their workday, providing valuable insights into their roles and tasks. The other two observations consisted of site visits to industrial manufacturing warehouses, allowing us to understand the driving environments that trucks must navigate. This provided valuable insights that were not captured during the interviews, as these did not involve on-ground workers. By taking notes, images and video recordings, a comprehensive understanding of the complexities associated with the implementation of an ABET was attained. This approach also facilitated a clearer perception of where it could fit in within the value chain.

4.2 Literature study

In order to gain a thorough understanding of the current research on technological developments in the logistics industry, a literature study was carried out. The current state of knowledge in the field was investigated, which involved identifying key research themes and trends, as well as any gaps or limitations within the existing literature. First, a general search based on the scope and pre-study was carried out through Google Search, Google Scholar and KTH Primo to gain a broad overview of the existing knowledge and research landscape related to automation. The search process included two methods: keyword searches and filtering, to find relevant information from different sources, such as academic journals, and reports from government agencies and industry associations. Initially, keywords directly related to the research topic were used, like 'automation', 'logistics industry', and 'customer involvement in value offerings'. From the initial findings, keywords such as 'socio technical shifts', 'radical innovation', 'business model innovation', and 'driver shortage' helped narrow the search. To make sure to investigate relevant topics, a trend analysis was conducted using the DHL trend radar (DHL, 2023), see Appendix A. This generated topics such as 'big data analytics', 'remote work & teleoperation', and 'supply chain diversification'. See Figure 4.2 for a full overview of the keywords explored. Filtering was then applied to limit the search results to published papers from the last five years to ensure that the information obtained was recent and up-to-date, given the rapid development of autonomous technology (Sindi and Woodman, 2021).

The information gathered from each source was categorised into clusters based on subject matter and content inspired by the thematic network analysis by Attride-Stirling (2001). This process facilitated knowledge sharing, and by using a systematic approach, 80 articles and reports could be analysed. First, a high-level theme was set to identify broad clusters based on shared characteristics across the data set. This provided an overview and helped in understanding the general patterns presented in the data. Once the high-level themes were established, each cluster was refined further into smaller sub-themes that exhibited specific and defined characteristics. This fine-tuning process involved delving deeper into each cluster to identify common issues, concerns or topics being discussed. Finally, the sub-themes were examined to determine similarities and contradictions within and across them. By doing so, valuable insights was gained, revealing the nuanced

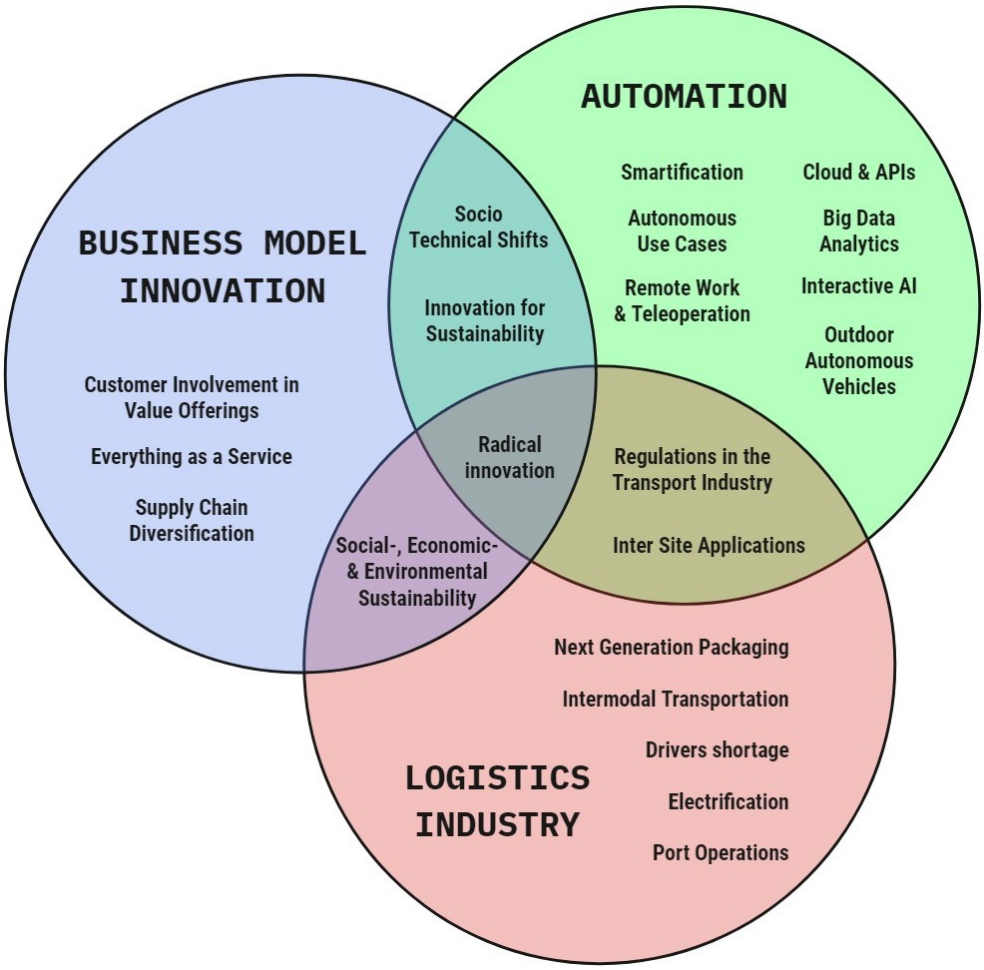


Figure 4.2: An overview of the literature study, where all areas explored are displayed and how they interact with each other.

perspectives and complexities present in the data.

4.3 Semi-structured interviews

To gather information from the industry on the market's view on autonomous vehicles and where they could potentially fit in, semi-structured interviews (Wikberg-Nilsson et al., 2015) were conducted with 10 industry professionals. As the aim of the interviews was to gain a deeper understanding of customer needs in the autonomous freight industry, several different key stakeholders with experience and knowledge in this field were approached, see table 4.1 for an overview of the respondents and interview duration. All respondents

have been anatomised in this report to enable all respondents to freely express their thoughts without any reservations or concerns about being identified. This allowed for open and honest responses that might have otherwise been withheld.

#	Title	Industry	Duration
1	Head of Logistics A	Industrial	52 min
2	Head of Logistics B	Fast moving consumer goods	53 min
3	Head of Strategic Logistics Development and Innovation	Port	61 min
4	Logistics Innovation Manager	Logistics Service Provider	60 min
5	Logistics Manager A	Fast moving consumer goods	55 min
6	Logistics Manager B	Fast moving consumer goods	55 min
7	Logistics Manager C	Fast moving consumer goods	55 min
8	Logistics Manager D	Industrial	53 min
9	Senior Logistics Business Consultant	Logistics expert	55 min
10	Site manager	Fast moving consumer goods	48 min

Table 4.1: Interviewee Respondents

Experts were approached for their in-depth knowledge of the logistics industry, where two of the experts (respondents 3 and 4) came from the port industry and the third (respondent 9) had experience with investigating the implementation of self-driving trucks. These interviews offered unique perspectives on the complexities of the logistics industry as a whole, focusing not only on customers' expressed needs but also on what industry insiders perceive as customers' actual demands and the inherent challenges involved. This was crucial since individuals might express one thing while behaving differently. The selection of ports was made based on the assessment that they are the most probable locations for the introduction of autonomous trucks.

The other interviewees were chosen because of their different roles and functions based on the findings from the pre-study. These interviews provided insight into real customer experiences to ensure a customer-centric approach.

In order to reach potential interviewees, a message was sent out on LinkedIn targeting individuals with job titles related to "logistics", such as transport planner, logistics developer/manager, and supply chain innovator. Furthermore, the pre-study participants' contact networks was utilised to identify individuals involved in the port industry. A questionnaire was sent out to the members of a European shipper association, however this did not generate any interviews.

To provide a framework for the interviewer and ensure important questions were asked, an interview guide was created, see Appendix B, to help stay focused on the research topic. The guide provided consistency across all interviews by asking all participants the same core questions, while still allowing for flexibility in the conversation to tailor follow-up questions to their specific responses. The first part of the interview, see Figure 4.3, addressed current daily operations, the transport need and the challenges the companies are facing today.

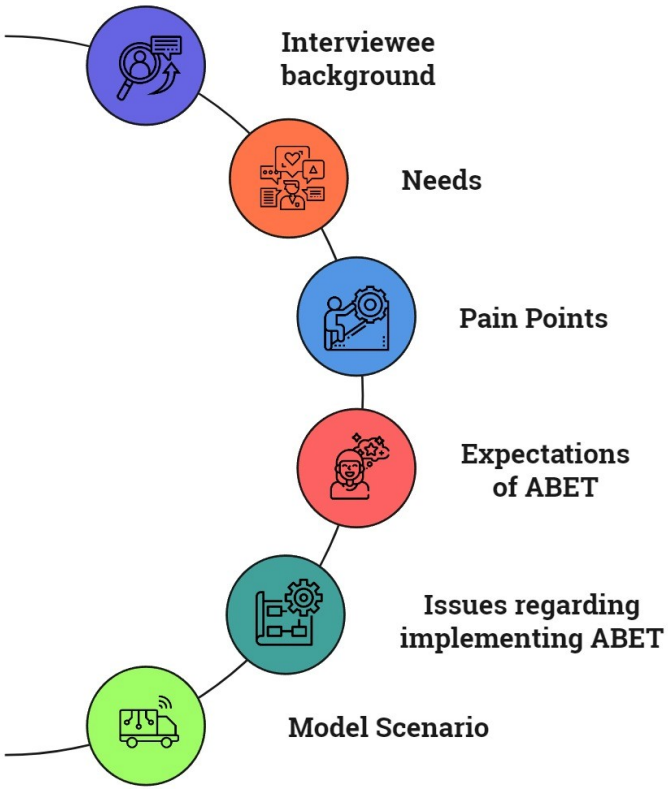


Figure 4.3: Interview guide process.

The second part of the interview focused on automation, including the interviewee's perception of it. In the third part of the interview, a model scenario was discussed. The model scenario was built to provide a foundation to make sure that personal knowledge of automation was not the basis of discussion. The respondent were asked to describe how they would approach the situation and what benefits and challenges they could find with the given solution. By presenting this common scenario to all interviewees, the responses could be compared and common themes or patterns were identified.

To test and refine the interview guide before conducting the full study, the guide underwent a refinement process after the first interview. By doing so, any problems or issues with the interview guide was identified and solved, such as unclear or confusing questions and the length of the interview. While certain formulations were modified, the overall structure and content of the guide remained unchanged.

Each interview had a duration of approximately one hour and took place online, where all interviews were recorded using the integrated software of the online meeting platform. During each interview, one researcher conducted the interview and asked questions from the interview guide, while the other researcher took notes on significant topics that required further exploration. Each interview was transcribed using a transcription software to ensure accurate representation of the participants' responses. Subsequently, the transcriptions were coded inspired by open coding (Corbin and Strauss, 1990). All transcriptions were read through, where codes were assigned to different segments of the interview. These codes represented concepts, ideas or themes that emerged from the data. It allowed capturing a wide range of ideas and perspectives expressed in the interviews. This method utilised inductive reasoning, allowing for new codes to emerge from the data rather than imposing preconceived categories. The coded data was then organised and analysed to uncover connections, relationships and insights, similar to the pre-study.

After consolidating the findings, a thorough analysis was conducted to determine the prerequisites necessary for a LSP to implement an ABET solution. The recurring themes, patterns and concepts that emerged from the interview data were carefully reviewed and compared with the literature review, thereby identifying the implications of the results and their alignment with established theories. This process enabled the systematic analysis of

the interview data to formulate theories and explanations rooted in the data itself. Moreover, it was crucial to comprehend and interpret the subjective experiences of the interviewees, with the goal of identifying their unique meanings and interpretations. This aimed to gain insights into specific knowledge that could be applied to multiple segments beyond those initially stated, ensuring comprehensive coverage of all relevant aspects.

4.3.1 Model Scenario

The model scenario for the interview guide relied on information gathered from the pre-study interviews and literature study. The focus of the discussion was not on delving into the technical intricacies of the technology itself. Instead, the objective was to gain an understanding of the customers' preferences regarding the functioning of the operations. Therefore, the degree of automation was set to SAE level 4 (high driver automation), where the system is capable of responding to changing conditions. However, it still requires a human operator to be present and vigilant, and therefore is not fully autonomous (SAE International, 2021). The context in which the truck operates is inter-site applications, characterised by the presence of fenced areas that are designed to be accessible only to authorised personnel. This creates a controllable environment and limits regulatory oversight. Nevertheless, it should be emphasised that the truck will be able to drive on public roads between these fenced areas, and will not be limited to operating solely within them. The distance between locations is not a matter of concern within the scope of this study, however it is important to note that because of the context restriction, the scenario is most suitable for nearby delivery, where the transport requirement is low-speed and enables inter-site transportation. Due to this, last-mile delivery has been excluded from the study due to its complex flow. Furthermore, it is assumed that issues such as charging infrastructure is functioning properly.

4.4 Creation of Business Case

The team engaged in extensive discussions to further understand the possible customer value of the introduction of autonomous trucks. By sharing perspectives, insights and interpretations of the data, a collaborative and reflexive analysis process was fostered. This approach aimed to cultivate a

deeper understanding of the data by capitalising on the collective expertise and diverse viewpoints of the research team. It facilitated in-depth discussions, critical reflections, and the generation of novel insights and interpretations through the synthesis of individual researchers' perspectives. This were then synthesised into recommendations for the industry. To further enhance this process and establish a connection with the existing literature, a business model case was constructed, with a particular focus on the value proposition. This involved drawing inspiration from the tool Value Proposition Canvas (Osterwalder et al., 2014), see Figure 4.4, where a customer profile was summarised and a value map was created, including the offering an ABET could provide.

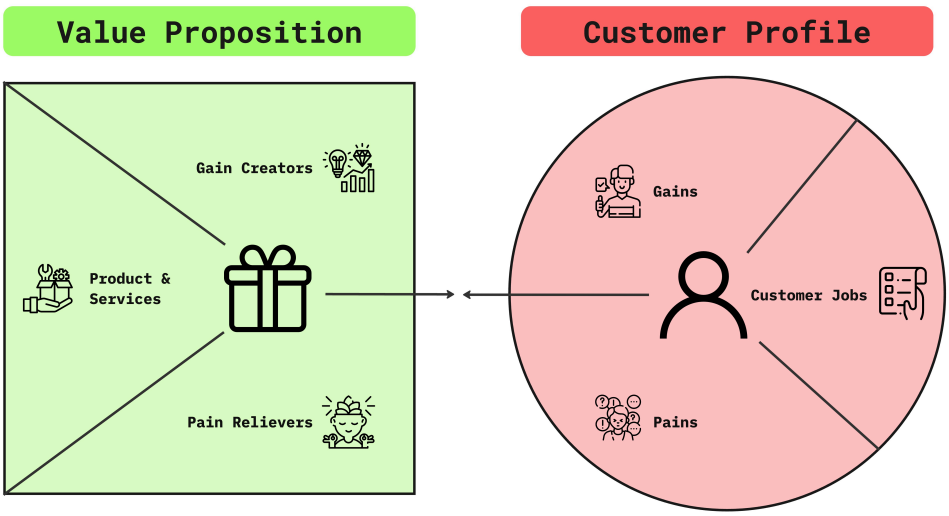


Figure 4.4: Value proposition canvas.

By adopting this approach, the focus was narrowed down to the specific aspect of the business model that centres around the customer's point of view.

Chapter 5

Results

This chapter provides a thematic analysis of the interview responses addressing the fulfilment of customer needs by ABETs (Autonomous Battery Electric Trucks), the challenges related to implementing ABET, and the prerequisites required by a logistics service provider LSP to operate an ABET solution. The result is structured around the main themes that emerged from the interviews.

5.1 Transport Logistics Status Quo

Today's transport logistics industry is a complex and heterogeneous matrix of operations involving multiple stakeholders, communication channels and transport methods. In large, it could be narrowed down into four main activities: Transport planning, transport execution, goods handling and goods tracking. Various stakeholders independently and interdependently perform these actions concurrently.

5.1.1 The Crucial Role of Communication, Coordination & Vendor Networks in Logistics Operations

The logistics industry is known for its diverse transport strategies, often involving vendor networks. One Logistics Manager mentioned that third-party distributors handle final sorting and distributing of goods across their network to reach the final destination. These distributors also manage middle-mile flows, showing how external partners play an integral role in the

logistics chain. Another Logistics Manager further highlighted the use of third-party vendors for transport between warehouse hubs, demonstrating the extensive reliance on vendor networks in transport strategies.

Communication and coordination play a central role in logistics. The Site Manager shared that drivers and warehouse personnel utilise verbal communication to determine the correct docking ports for loading and unloading. This aspect of communication is essential to maintain safety and correctness during the operation. Meanwhile, a Logistics Manager noted that interaction with transport service providers mostly occurs via traditional channels such as email and phone. Despite some parts of the industry embracing digital transformation, such as digital booking, these manual methods of communication remain a cornerstone of the day-to-day running of logistics operations.

Shippers shoulder substantial responsibility, as highlighted by the FMCG Head of Logistics. Shippers are expected to track their goods, often dedicating human resources specifically for this purpose. The FMCG Logistics Manager revealed a long-term ambition within the industry to gain more control by owning their own fleet. This control also extends to managing inbound deliveries, with a desire to optimise volumes and avoid congestions, which require a deep level of involvement in the process.

> *"Have 3 full-time employees just to sit in the system and track where everything is."*
> - Head of Logistics in FMCG industry

5.1.2 Transportation Acquisition and Prioritisation: Factors, Collaboration & Differentiators

The process of acquiring transportation services involves various factors and considerations, as highlighted by different respondents. The Logistics Manager explained that for different projects, they approach the market and request solutions, ultimately selecting one vendor with whom they place bookings. The Logistics Innovation Manager noted that logistics service providers (LSPs) prioritise vendors based on performance metrics such as on-time and complete delivery and pick-up, as well as their commitment to zero emissions. The decision-making process is often collaborative, involving central procurement, logistics teams and local stakeholders, as mentioned

by the Site Manager. The Site Manager further emphasised that there is a dedicated central team responsible for procurement from specific suppliers. Additionally, one Head of Logistics revealed that central teams issue tenders and secure partners for long-term contracts. The decision-making process for vendor selection may involve regional analysis based on vendor maturity. Moreover, data-driven approaches are becoming more prevalent in purchasing decisions highlighted a shift toward strategic decisions based on current needs and data analysis.

When it comes to priorities in transportation acquisition, cost and transit time are often considered essential, with reliability also influencing choices, according to the Logistics Managers. Tenders are evaluated quantitatively in terms of costs and terms, but also qualitatively based on the value they bring to customers. Balancing service level and cost is crucial, with factors such as sustainability, customer service and the appropriate type of vehicle taken into account, as described by the industrial Head of Logistics. Logistics Manager B has highlighted the importance of long-term investment in electric vehicles, viewing them as a potential source of competitive advantage, with both cost and sustainability in mind. Equally important, the FMCG Logistics Managers stressed that swift delivery times are critical to fulfilling customer expectations.

Reliability emerged as a crucial factor in tendering and purchasing decisions, according to Logistics Managers and the Logistics Innovation Manager. All respondents emphasized the importance of ensuring goods are picked up and delivered on time and in full. Trust in the transport solution is paramount. Vendors need to demonstrate their ability to meet these expectations, as reliability is a key need expressed by shippers. However, challenges arise in terms of capacity, given that the logistics industry predominantly comprises numerous small carriers, as noted by the Logistics Innovation Manager. Logistics managers strive to optimize delivery time, costs, and the number of pick-ups from the warehouse to enhance reliability, while acknowledging the increasing importance of environmental considerations.
While sustainability is not typically offered as a differentiating service provided by LSPs, it is gaining relevance in tendering and purchasing decisions. Logistics managers indicated that customers prioritise low-cost, convenient, and timely deliveries, with sustainability coming into play when comparing equally viable options. Although sustainability is not the primary focus, it is increasingly recognised as an important factor to consider. The

ability to provide data and operational efficiency serves as a competitive advantage for vendors, aligning with the growing demand for sustainable practices. As sustainability continues to gain traction, Logistics Managers and the Logistics Innovation Manager highlightedthe need for environmental aspects to be taken into account alongside traditional priorities such as cost and convenience in order to meet evolving customer expectations and remain competitive in the market.

5.1.3 Driving Sustainability in the Logistics Industry

Insights from the industry expert highlighted the importance of sustainability in the logistics industry, leading to a significant push in this direction. The Logistics Innovation Manager stated that meeting science-based targets[1] necessitates investments in zero emissions or low zero emissions alternatives, leaving little room for non-sustainable options. This drive for sustainability was driven by management, indicating a top-down approach to promoting environmentally friendly practices, as stated by some Logistics Managers. Shippers actively invest in costlier sustainable solutions not only to align with sustainability goals but also to gain valuable experience with these alternatives, as noted by the Senior Logistics Business Consultant. Additionally, Logistics Manager A emphasised the significance of social sustainability, highlighting the importance of avoiding partnerships with distributors and carriers that had poor working conditions.

In their efforts to align with sustainability initiatives, shippers prioritised the reduction of emissions and improvement of efficiency. Logistics Manager A revealed that optimising transport volumes and fill rates was a priority to enhance efficiency and lower costs. They aim to minimise "air" in shipments and fill trucks as efficiently as possible, as stated by Logistics Manager D. Shippers also exert pressure on their distributors to prioritise environmentally friendly practices, highlighting the growing importance of sustainability, according to Logistics Manager A. The Logistics Innovation Manager further emphasised the need for companies to invest exclusively in zero emissions alternatives as a means of reducing their environmental impact, in line with the Science Based Targets Initiative.

[1]Targets are considered ‘science-based’ if they are in line with what the latest climate science deems necessary to meet the goals of the Paris Agreement – limiting global warming to well-below 2°C above pre-industrial levels and pursuing efforts to limit warming to 1.5°C.

5.1.4 Navigating the Challenges & Opportunities of Electric Truck Logistics

The electrification of truck logistics presents both challenges and opportunities, as indicated by insights from various respondents. Electric trucks, despite their long-term investment potential, have yet to yield returns, as noted by Logistics Manager B. Cost considerations play a crucial role, given that electric trucks are more expensive and require optimal utilisation to justify the investment, as highlighted by Logistics Manager B.However, the Logistics Innovation Manager expressed the expectation that BETs will achieve price comparability by 2026, suggesting a positive cost trend for sustainable transportation, according to the Logistics Innovation Manager.

Operational implications of electrification are observed, with certain challenges becoming evident. The charging time for electric long-haul trucks is a notable challenge, as highlighted by Logistics Manager C. The overall cost of ownership is significantly influenced by driving and operational efficiency, with potential variations of up to two times depending on driving conditions, as stated by the Logistics Innovation Manager. Human operating hours play a crucial role in balancing the differences between Internal Combustion Engine vehicles and BETs, especially in long-haul operations, according to the Logistics Innovation Manager, due to limiting factor of driver rest time. The transition to electric trucks necessitates adjustments in driver behaviour and planning, as indicated by Logistics Manager B.

> "*Regardless of the vehicle, the driver still has to stop after four and a half hours and have a 45 minute break, which if you have megawatt charging at site, you can recharge your battery to at least 80% in 45 minutes, be back on the road for another four and a half hours, and then no driver's allowed to drive more than nine hours in a day in Europe.*"
> - Innovation Manager at LSP

The shift to electric trucks is primarily driven by sustainability concerns. The Logistics Innovation Manager attributed the decreasing abatement cost of electric vehicles to various factors, including lower electricity prices, improved operations, economies of scale, and the rising costs of traditional Internal Combustion Engine (ICE) trucks. Despite cost concerns, some respondents have successfully electrified their deliveries without incurring additional expenses on their end, as stated by the industrial Head of Logistics.

However, the increased asset cost of electric trucks creates challenges as it requires increased asset utilisation compared to the otherwise "cheap" ICE trucks. Similar to other expensive assets such as ferries and planes it would need to be in constant use to be cost comparative, according to the Port expert. Battery swapping is seen as a solution to mitigate charging times and enable electric trucks to operate at least similar to ICE trucks, as emphasised by one of the Head of Logistics.
The Logistics Managers also noted challenges with implementing BETs. The charging infrastructure and longer charging times impact the flow of operations. Technical issues with electric trucks are still being resolved, and the transition requires a new way of working and planning. Also, the perception of battery electric vehicles as environmentally beneficial depends on the source of electricity, highlighting the importance of utilising "green" electricity to maximise sustainability, according to the industrial Head of Logistics.

The market offering for zero-emission transportation is currently limited, leading companies to consider the available options and set "realistic requirements", as cited by a FMCG Logistics Manager. Another FMCG Logistics Manager noted that BETs is providing a competitive advantage in terms of sustainability and having the potential to lead to increased business opportunities. It was also highlighted that the focus on electric trucks is driven by public perception and the anticipation of future growth in this sector. According to the LSP innovation manager, some logistics service providers aim to have ownership over a portion of their zero-emission vehicles while relying on vendors for the remainder. This approach is partly influenced by the slow vendor adaptation and their inability to provide the new technology quickly enough.

5.1.5 Challenges & Pain Points Faced by Shippers & Transport Acquirers

Head of Logistics A mentioned that the current transport offering on the market makes it challenging to meet customer demands, such as 30-minute delivery time slots. However, the Senior Logistics Business Consultant highlighted that despite initial complaints, logistics professionals manage to arrange everything by the end of the week, even though the process often deviates from the initial plan.

> They are complaining quite a lot, but once the week has gone, everything has been arranged - they fixed everything. They actually have no clue how they did it, but in the end everything has been organised and that's the general characteristic of logistics. It's never going according to plan. So they are just always changing plans, just taking the situation as it comes.
> - Senior Logistics Business Consultant

The Logistics Managers highlighted inefficiencies at the site, specifically pointing out language barriers resulting from non-regular deliveries with unfamiliar carriers. These language barriers, stemming from drivers from multiple countries speaking different languages, can lead to inefficient actions and challenges during the logistics process. Examples of these inefficiencies include difficulties in understanding and following instructions, confusion during loading and unloading procedures, and potential safety hazards due to miscommunication.

The Head of Strategic Logistics Development and Innovation highlighted the challenge of unreliable estimated time of arrival (ETA) for trucks, leading to the need for standby personnel and causing waiting times during peak hours. Additionally, congestion at sites and insufficient information about truck arrivals or departures were mentioned as pain points by the Senior Logistics Business Consultant.

> "*..they [shippers] only know that the truck will pick up goods on Friday. Whether it will be 9 o'clock in the morning or six in the afternoon, no one knows.*"
> - Senior Logistics Business Consultant

The Logistics Managers highlighted that damage to goods often occurs in handling areas during loading and unloading, which is a significant pain point in logistics operations. The FMCG Head of Logistics also mentioned the struggle to maintain the right temperature for cold goods during these handlings, particularly in intermodal[2] transports. This is especially challenging considering that rail transportation is seen as a way to address the increasing driver shortage issues in Europe.

[2]Intermodal shipping refers to moving freight by two or more modes of transportation.

5.1.6 Challenges & Costs Associated with Driver Shortage in the Shipping & Transport Industry

Challenges surrounding driver shortage have profound implications for shippers and transport buyers, mostly manifesting as escalating transport costs and operational disruptions. During the COVID-19 pandemic, the surge in sickness-induced absence was a significant driver of cost increases, as pointed out by a Logistics Manager. Although prices have since stabilised, the persistent shortage of drivers continues to exert upward pressure on costs. It is highlighted from both the Logistics Managers and Head of Logistics that, from a shipper's perspective, costs are the primary consequence of the driver shortage. In contrast, third-party providers view it as a procurement complication.

The Head of Logistics A underscored the sizeable portion of operational costs attributed to salaries, with sickness and work-related injuries accounting for 10% and 6%, respectively. With automation touted as a potential solution to mitigate these costs, there are cautionary notes to consider. The Logistics Innovation Manager advised that the introduction of autonomous trucks should not merely be a labour cost-saving exercise. The goal should be broader, including the potential to run operations around the clock, enhance efficiency, decrease energy use, increased and other intangible benefits that this technological shift could realise.

The FMCG Head of Logistics identified multiple contributing factors to the driver shortage and increasing transport cost, highlighting the significant role of global events such as the Russian/Ukraine war. However, social factors are also at play. There is a noticeable dwindling interest in driving careers, with those currently in the profession gravitating towards e-commerce companies due to their shorter routes, higher remuneration and more appealing working hours. To counter these challenges, the EU has introduced legislation aimed at bettering the work conditions of non-local drivers. While this may improve social conditions, the FMCG Head of Logistics further noted that it also brings increased administrative overhead, subsequently escalating overall transport costs. Thus, for shippers, the driver shortage issue continues to be a significant cost driver.

5.1.7 Digital Tools in Truck Logistics: Pain Points & Aspirations for Integration & Visibility

The role of digital tools in current truck logistics is a topic of significant interest and discussion among industry professionals. According to the industrial Logistics Manager, warehouses often rely on multiple support systems that need to be integrated for a seamless digital experience. This integration challenge was recognised by various stakeholders, including Head of Strategic Logistics Development and Innovation, who emphasises the need to connect different systems and share information to reduce waiting times and improve utilisation, and efficiency, ultimately leading to reduced climate impact.

Digitisation has the potential to revolutionise tracking and visibility in truck logistics. The Logistics Managers highlighted the benefits of real-time tracking, such as improved customer satisfaction and the ability to plan for potential delays. However, challenges arise when it comes to stakeholders' willingness to share information and foster a collaborative mindset. As stated by Head of Strategic Logistics Development and Innovation, it is not always obvious for everyone to share information in order to have a transparent view of the entire transportation process.

The digitisation of waybills is a common pain point and aspiration within the industry. Logistics managers express the desire to eliminate paper waybills entirely and transition to digital alternatives. However, they have concerns regarding the manual processes currently in place and the continuous data entry on these papers, including recording damages and ownership status. Additionally, it is stated that a backup system should be in place in case of system crashes since operations cannot afford to halt.

5.2 Unlocking Value: The Potential of Autonomous Battery Electric Trucks

The implementation of ABET systems holds significant value for shippers and transport buyers, as indicated by the respondents. The industrial Head of Logistics emphasises that autonomous night driving would bring social value by eliminating the need for drivers to work during nighttime hours.

The Logistics Innovation Manager highlighted that ABETs could increase efficiency, enable round-the-clock operations, and overcome limitations imposed by driver operating hours and enable electric long-haul, a sentiment also echoed by the Senior Logistics Business Consultant. Within the FMCG segment, ABETs could optimise transport flows by removing the human factor, enhance cargo safety and enable more frequent pick-ups by removing the driver cost, thereby reducing delivery times.

Additionally, the industrial Logistics Manager pointed out the environmental sustainability aspect of ABETs, as autonomous and electric vehicles contribute to reduced emissions. The Senior Logistics Business Consultant further noted that if autonomous trucks produce zero tailpipe emissions, it would diminish the argument in favour of traditional railway transport.

5.2.1 Initial Deployment of Autonomous Battery Electric Trucks

ABETmight have significant potential in different areas of logistics operations. The Industrial Logistics Manager highlighted their suitability for port and railway environments, particularly for middle-mile deliveries from ports to nearby factories. This perspective aligns with the FMCG Head of Logistics, who expressed interest in establishing warehouses near railways and utilising ABETs to facilitate shuttle services between rail terminals and production facilities. Internal logistics flows also present a promising use case for ABETs. The industrial Head of Logistics suggested employing ABETs for milk rounds within industrial parks, and a FMCG Logistics Manager emphasised their feasibility in repetitive, internal transport operations. Furthermore, ABETs could effectively serve restricted areas during nighttime deliveries, as pointed out by another FMCG Logistics Manager.

While the value of ABETs is evident in long haul applications, the outlook remains more complex. According to the Senior Logistics, the ultimate goal is long-haul transportation, but the technology needs to be validated in multiple short haul use cases to demonstrate its worth. The Head of Strategic Logistics Development and Innovation similarly recommended focusing on short haul routes with repetitive tasks and minimal partners as a starting point. Additionally, an FMCG Logistics Manager saw the greatest potential for ABET in internal hub-to-hub routes, where warehouse personnel are available on both sides, reducing the need for drivers. It was also highlighted that

ABET brings value in long-haul scenarios where drivers otherwise need to be switched or take the mandatory rest time, making it easier to enable market presence and expansion to other markets as delivery times become less of an issue. The Logistics Innovation Manager further that ABETs could be key to unlocking zero-emissions long-haul transport, considering the lack of sleeper cabins in current BETs. Furthermore, the Logistics Innovation Manager emphasized that ABETs could be instrumental in unlocking zero-emissions long-haul transport, particularly due to the absence of sleeper cabins in current BETs.

> "*With autonomous vehicles, we could transport the cargo to our facility in Germany much faster, bypassing the driving limit constraints. There would be fewer stops and no need for reloading. We could achieve a delivery time comparable to competitors with a stronger local presence in Germany, starting at 4 p.m. and arriving by 8 a.m."*
> - FMCG Logistics Manager

Despite the promising applications of ABETs, logistics managers acknowledged the challenges involved in implementing them in certain scenarios. One of the challenges identified was the customer-facing role of drivers, as not all customers are willing or able to unload cargo themselves. Additionally, when trucks are required to make deliveries to multiple customers in a single route, implementing ABETs may not be feasible. As a result, it was suggested that ABETs should be deployed in flows where the driver's presence does not add extra value, particularly in cases where they are not responsible for loading and unloading. However, a FMCG Logistics Manager recognised the potential value of ABETs in improving delivery times for e-commerce companies. Specifically, if ABETs could operate between a warehouse and a distribution centre, delivering goods around the clock, it could enable next-day delivery for orders placed late at night.

5.3 The Complexities of Autonomous Truck Implementation & Infrastructure

The Logistics Innovation Manager raised a concern regarding the potential need for substantial infrastructural adjustments, such as the integration of connected traffic signals, which could incur high costs. This concern was

echoed by the Head of Strategic Logistics Development and Innovation, who highlighted the financial burden that existing ports may face in making the necessary investments to facilitate ABETs. However, it was also noted that small changes, such as allocating specific areas within terminals for the vehicles, could address safety and reduce traffic complexity. On a different note, a FMCG Logistics Manager expressed confidence in their ability to adapt their sites or operations to accommodate an autonomous solution if the need arises.

In the discussion of ABETs, ownership and control are two intertwined topics. The Logistics Innovation Manager emphasised that the implementation of ABETs has a significant impact on the entire transport system, leading LSPs to anticipate a future where they own more of the assets as they integrate autonomous solutions. In terms of fleet control, the Senior Logistics Business Consultant highlighted the necessity of a holistic approach to autonomous planning—one that involves several companies and is deeply integrated with the physical internet and hardware, reflecting a control tower scenario such as in the flight industry.

Regulatory hurdles and safety requirements are other critical considerations when it comes to ABETs. The Senior Logistics Business Consultant emphasised the stringent nature of the approval process for testing autonomous solutions on public roads, thereby suggesting controlled environments like yards for initial pilots. Furthermore, the inherent unpredictability of human drivers was cited by the Logistics Innovation Manager as a significant challenge for autonomous trucking. Echoing this safety-first mindset, most respondents thought that piloting autonomous trucks in controlled environments with a remote driver while acknowledging the need for further technological development is the way forward.

However, the question of who should take the lead in implementing ABETs brought out differing opinions. The Logistics Innovation Manager suggested a collaborative effort between transport providers and shippers for deploying autonomous solutions, while the Head of Strategic Logistics Development and Innovation proposed that transport companies, which stand to gain significantly from improved efficiency, should primarily drive the push for automation. Either way, all logistics managers stated that if they were to start shipping with autonomous technology, it would be up to the transport providers to suggest and implement such a solution.

> "*It is for their own gain. There is nothing else to it. It is fine that they want to optimise their flows - there is nothing strange about that - but that's how it works. It's not us who set the demands on them, it is they themselves who increase the efficiency of their flows.*"
> - Head of Strategic Logistics Development and Innovation

The issue of investment also arose during these discussions, with the industrial Head of Logistics underlining the necessity for a large player to invest heavily to make autonomous solutions efficient and affordable. However, the question of who that player might be was left open-ended.

> "*You need somebody to go big, invest in it to make it efficient to bring the price down and it's who's going to be first to market to do that.*"
> - Industrial Head of Logistics

Despite these challenges and complexities, the Logistics Innovation Manager suggested a gradual, hands-on approach to implementing ABETs, including conducting pilot cases to gently introduce people to the technology and allow them to learn by doing.

5.3.1 Assessing the Investment Implications

Implementing ABETs comes with its challenges, as voiced by several respondents. The Logistics Innovation Manager warned about the upfront cost and the potential mismatch between customer willingness to pay and the company's eagerness to innovate. The FMCG Head of Logistics highlights the potential operational challenges of adopting ABETs, particularly if it requires a logistics network rearrangement by bringing warehouses closer to customers, which could have a negative impact on the business.

> *"Your customers willingness to pay is not going to match your willingness to innovate and do the right thing by the industry.*"
> - Logistics Innovation Manager

As logistics sites prepare for electric operations, the Logistics Innovation Manager advised that preparations should also be made for autonomous vehicles. However, it is crucial to consider an important factor: the contract

logistics industry is highly cost-sensitive, and significant investments in autonomous vehicles will only be made if there are clear benefits and efficiency improvements. The Senior Logistics Business Consultant, in line with this concern, observed a disconnect between the concept designs of autonomous vehicles and the actual needs of customers, making it challenging to evaluate the business case for ABET. Furthermore, they highlighted a disparity between what OEMs claim their automated vehicles can do and the detailed requirements that surface when discussing with shippers. This disparity raises potential conflicts during in-depth discussions and calls for careful consideration of the practical aspects of integrating autonomous vehicles into the logistics industry.

> "*...Infrastructure requirements for autonomous are not very well understood. And the kind of push-backs that I had from contract logistics was that contract logistics is already a very cost sensitive environment. And even when it comes to changing a light bulb, they want to know how much is that going to cost me?*"
> - Logistics Innovation Manger

However, the implementation of ABET is not without its merits. According to the FMCG Head of Logistics, adopting autonomous operations at scale under the right conditions could potentially yield the operational benefits associated with ABET. It is crucial, though, to strike a careful balance during the transition from pilot projects to full-scale operations to ensure that short-term efficiency is not compromised. From a practical standpoint, the Head of Strategic Logistics Development and Innovation emphasises the need for ABET to closely resemble regular trucks as much as possible. This is due to the limitations in making significant adjustments for a small fleet of autonomous trucks. By aligning ABET with the existing infrastructure and operations, the integration process can be smoother and more practical.

> "*Think about it yourself - if you get offered something new, and you do not know what to do with it, and to make it work it might cost you a lot of money. You will also think "Oh I don't know, lets wait and See"*"
> - Senior Logistics Business Consultant

When it comes to decision-making, the FMCG Head of Logistics provided a practical perspective: easy-to-implement options are those that are cost-neutral or offer a payback within four years, without negatively

impacting operations. Echoing this sentiment, the industrial Head of Logistics stated that cost will be the ultimate determinant in the decision to invest in autonomous transport, considering both the payback time and the lifespan of the investment.

5.4 Operational impact of Autonomous Trucks

As the transition to autonomous trucks progresses, the respondents offered their insights into how they envision booking and managing these vehicles operationally.

5.4.1 Digital Management of Autonomous Trucks: Insights, Expectations and Challenges

The respondents expressed no concerns about managing the autonomous truck digitally, viewing it as a non-issue. The Logistics Innovation Manager pointed out that their LSP already schedules and manages all their vendors digitally, suggesting that extending the same approach to autonomous trucks would be a natural progression.

The Logistics Managers imagines a future where autonomous trucks appear as needed, without anyone having to call for them. However, it was raised that the system need to be adaptable. If there are last-minute changes to the goods, there should be an efficient way to update this in the system, like a screen with selectable options.

A FMCG Logistics Manager expressed a desire for maintaining control over the remote control centre in the event of implementing autonomous trucks. This indicating that while embracing the autonomous feature, it is till important to have some level of manual oversight. In terms of system integration, the respondents discussed the need for linking two systems: the Transport Management System and the Warehouse Management System into the digital ecosystem.

Several challenges were raised during the discussion. Handling waybills was specifically pointed out as an issue. Currently, the driver is responsible

for taking the signed waybill back to the supplier, which would require digitisation on both ends if the waybill is to be eliminated. Additionally, FMCG Head of Logistics highlighted the necessity of clearly transfer ownership and responsibility, which could pose a challenge in a driver-less setup. However, it was also acknowledged that this technically could be done electronically.

5.4.2 Operational Challenges & Potential Solutions for Arrival, Departure & Dock Interaction of Autonomous Trucks

Several respondents envisioned the use of cutting-edge technologies to assist with loading and unloading pallets of goods onto the ABETs. Some logistics managers saw no issue with warehouse workers handling the unloading and loading of goods in the case of driver-less trucks, as long as it is planned in advance. Ultimately, the decision revolves around cost and resource considerations. However, it was emphasised that there must be a way to guarantee secure loading of goods to ensure safe driving. Additionally, there is a need to clearly define who is liable in case of any accidents involving the goods, as currently, the responsibility lies with the drivers. Although, the respondents showed little interest in taking on a larger economic liability and responsibility of the goods during transport.

Concerns were raised about how warehouse personnel will communicate with the driver-less trucks. The Site Manager and a few Logistics Managers questioned how personnel would know what the truck wants or needs and how to ensure personnel take responsibility for the truck and its loading. In response to these concerns, an app was envisioned by a Logistics Manager, which could also be used to communicate and report late deliveries. In addition, Head of Strategic Logistics Development and Innovation emphasised that an autonomous system must be able to communicate with all operators and stakeholders.

Arrival and departure procedures were discussed, and some logistics managers expressed uncertainty regarding the actions the truck should take upon arriving at the gate. The Head of Strategic Logistics Development and Innovation suggested that the truck should be able to register its arrival and park in an assigned spot. The Site Manager acknowledged that timing demands could

pose challenges during arrival and departure. However, a potential solution was proposed by the FMCG Head of Logistics, suggesting the implementation of pre-announced arrivals and license plate scanning. Another question raised was the issue of truck control while on-site. Logistics managers questioned who should have control over the truck and how this control should be established. It was also emphasised that sites have different shapes and layouts, emphasising the need for adaptability in ABETs.

5.4.3 Operational Expectations & Challenges for Autonomous Trucking Solutions: Service Level

The respondents highlighted the criticality of maintenance in autonomous solutions, noting that maintenance practices currently vary across different sites for their existing technical solutions, such as autonomous warehouse technology. Some sites rely on service agreements with hardware providers for their warehouse trucks, while others handle maintenance through their own personnel. To ensure a smooth transition, it was suggested establishing a similar arrangement for autonomous trucks. In addition, given the newness of the technology, it was believed that the transport provider would need a maintenance team available to address any potential issues.

The industrial Logistics Manager raised concerns about the potential challenges arising from a breakdown of an autonomous truck. It was highlighted the greater difficulty of replacing an ABET compared to a conventional diesel truck, given the higher availability of the latter. Taking a pragmatic view, both FMCG Logistics Manager and the industrial Head of Logistics emphasised the primary objective of ensuring the successful delivery of goods, regardless of the issues that may arise. They underlined the responsibility of transport companies to resolve delivery issues and bear the associated costs. Promptly fixing any breakdowns was seen as crucial to avoid disruptions in deliveries. It was further noted that transport companies typically have access to a large fleet of trucks, enabling swift replacements in case of breakdowns. The Site Manager echoed the need for timely deliveries and stressed the importance of having alternative solutions ready to step in if an ABET is not functioning properly.

On the topic of overseeing operations, the Logistics Innovation Manager emphasised the need for LSPs to have a way to oversee and manage all operations, even if they entrust the operation and trust of autonomous trucks

to a vendor.

5.5 Stakeholder Perceptions on the Feasibility of Autonomous Electric Trucks: Balancing Safety, Trust & Technological Progress

The respondents offered diverse views on the feasibility of implementing ABETs. The general consensus among respondents suggests that while ABETs are seen as potentially beneficial for the future of logistics, there are considerable challenges and uncertainties to be addressed before they can be widely adopted. Safety concerns appeared to be a dominant factor among the respondents, affecting their perceptions. All respondents expressed an excitement about autonomous technology but simultaneously voiced safety reservations. It was highlighted the necessity of establishing trust in autonomous systems before their wide-scale implementation.

There was a consensus on the need for extensive testing to ensure reliable performance and safety of autonomous vehicles. Additionally, the importance of changing public perception about autonomous vehicles was acknowledged, with the Innovation Manager emphasising the need to break the mindset of automatically blaming autonomous vehicles in case of an accident. Considering these factors, the FMCG Head of Logistics expressed the belief that the implementation of long-haul ABETs was still a distant reality. Similarly, a FMCG Logistics Manager did not foresee autonomous trucks operating independently in the near future. This sentiment was echoed by the industrial Logistics Manager, who recognised the potential of autonomous vehicles but stated that their organisation currently had no preparations for autonomous inter-site application.

Despite these concerns, a few respondents recognised the potential advantages of adopting autonomous technology. It was acknowledged that there would be a great potential first-mover advantage a LSP would have and highlighted how improved public perception can be achieved if autonomous vehicles enhance people's lives, reduce noise and mitigate risks. All respondents also expressed optimism, suggesting that once autonomous trucks are fully integrated and

functional, they could greatly benefit operations. However, both a FMCG Logistics Manager and the Site Manager stated that their current focus lies on electrification rather than autonomous technology, although they would be interested in exploring autonomous electric solutions in the future.

Chapter 6

Analysis

This chapter focuses on interpreting the results by comparing them to the literature study.

6.1 Key Considerations in the Transport Procurement Process

Transport acquirers select vendors based on their capability to effectively address their specific challenges and requirements. One such challenge, is the increase in demand for faster deliveries as pointed out by Sindi and Woodman (2021). Abrahamsson et al. (2023) and Flodén et al. (2017) emphasises the significance of cost as a critical factor in the procurement process. However, even though the interview findings confirm this observation, the offerings provided by vendors exhibit a notable similarity to one another, making the concept of cost-effectiveness multifaceted and extending beyond the selection of the cheapest option alone. Flodén et al. (2017) also raised other factors such as reliability as important in acquisition, which during the interviews, emerged as a significant factor, encompassing multiple aspects. These included the number of completed deliveries, ensuring goods remained undamaged, meeting on-time delivery expectations, and the frequency of transits along a specific route. According to Flodén et al. (2017), the significance of reliability appears to diminish for longer transit, as cost takes precedence over reliability in such cases. However, the interviews showed that acceptability of a longer duration is dependent on whether the transit time aligns with the desired service level for customers, like the ability to

offer next day delivery, which ultimately relate to cost.

6.2 Balancing Environmental Sustainability & Customer Demands

Shippers choose transportation modes based on their needs and requirements. While trains are preferred for capacity and predictability, trucks are consistently favoured due to their inherent flexibility, as indicated by Neuweiler et al. (2017). However, one interviewee emphasised the importance of sustainability and expressed a desire to reduce reliance on road transport. If road transportation achieves zero emissions, it could become more attractive and potentially replace other modes such as rail, as highlighted by another respondent. Though, while all respondents acknowledged sustainability as important, the majority felt that the industry's drive for sustainability stemmed primarily from societal demands rather than internal company incentives. For example, with sustainability goals, such as the European Commission incentives (2023) leaving little room for non-sustainable alternatives, the consideration of sustainable options has become a prerequisite rather than a mere preference when purchasing new transport solutions.

The literature highlights environmental sustainability as a key advantage of automation. While sustainability gains importance and provides a competitive edge, consumers' purchasing decisions are primarily driven by factors like low cost, convenience and fast deliveries, rather than a willingness to pay more for sustainable options. Therefore, the primary focus remains on optimising flow, maximising pickups, lowering costs, and ensuring timely deliveries, with sustainability playing a secondary role in decision-making processes. In other words, while the literature focuses on societal benefits, shippers prioritise their individual operational concerns. Consequently, from the perspective of a LSP, when marketing an ABET, it is crucial to highlight the benefits that have a direct impact on the shipper's operations.

Similarly, while researchers, for example Anderson et al. (2016) and Sen et al. (2017), primarily focus on the potential benefits of reducing congestion on public roads, shippers are more concerned with the congestion occurring at their own facilities. To address this issue and optimise the flow of goods while minimising inefficiencies and resource consumption, an expectation of

ABETs is nighttime driving. The FMCG industry, in particular, recognises the potential in utilising nighttime driving to generate outbound deliveries during the night, which can then be delivered to the end customer the following day.

Both the literature study and the interviews converged on key aspects concerning electrification. Firstly, the charging infrastructure and longer charging times are posing a threat to operational efficiency, as pointed out by Balasaygun (2023). Moreover, electric trucks incur higher costs compared to traditional trucks, posing a short-term investment challenge (Abrahamsson et al., 2023). Thus, it might not be feasible to replace an entire fleet with electric trucks, but rather use an incremental implementation. This goes in line with what Szekely and Strebel (2013) suggests for new innovations, and a parallel can be drawn to ABETs as well.

6.3 How Autonomous Trucks Address Industry Pain Points

The logistics industry faces numerous pain points that hinder its efficiency and ability to meet evolving customer demands. These challenges include difficulties in securing sufficient capacity, manual planning processes, limited innovation, a shortage of drivers resulting in rising costs, and a lack of transparency in vendor operations. However, amidst these challenges, autonomous trucks offer promising solutions to address these pain points and drive the industry towards a more efficient and sustainable future.

6.3.1 Embracing Digitisation & Supply Chain Visibility

For ABETs to operate effectively, they require the capability to collect and analyse data, not only for navigating their environment but also for managing orders and interacting with warehouse staff. While digitisation has made progress in areas like digital booking systems, many aspects of the industry still rely on analogue methods such as paper waybills. The respondents recognise the increasing importance of digitisation and the need for improved supply chain visibility in the future. For example, access to data is highly

valued as it enables better performance understanding and control over the flow of goods. Planning based on estimated time of arrival allows shippers to avoid standby personnel waiting for unexpected delays and reducing congestion at sites. Nevertheless, successful implementation of ABETs faces challenges related to system reliability, as highlighted by Monios and Bergqvist (2020), and stakeholder collaboration. Within a company's transport flows, multiple actors are involved, creating a barrier to sharing information with competitors. Therefore, it is important to offer strong data integrity.

6.3.2 Assessing the Driver's Role: Reducing Inefficiencies & Addressing Driver Shortage

ABETs offer significant value by eliminating human inefficiencies and reducing transport lead times according to Fagnant and Kockelman(2015) and Hashimy and Rosines (2021), aligning with the expectations of the interviewees. One particular pain point the shippers express is human error, which frequently results in delays. Additionally, damage often occurs during docking operations, and an automated system could be the solution as suggested by Flämig (2016). However, unless the loading and unloading processes are also automated, there is a risk of pushing the problem over to the shipper, forcing them to take responsibility of the goods staying intact. This was not seen as a big hinder for adoption by the interviewees, however it requires a different level of effort from the shipper as as goods personnel need to be trained to take over the responsibility from the driver. Nevertheless, there are significant benefits to removing the driver from operations, such as mitigating language barriers and reducing substantial costs associated with salaries and sick leave. Another critical aspect though is the shortage of drivers, as highlighted by Chen and Lu (2021) and Sindi and Woodman (2021). Even if shippers may not directly experience the driver shortage, they still face increased costs as an indirect consequence. By eliminating the need for drivers or having a single driver operate multiple vehicles, this issue can be effectively addressed.

6.3.3 Enhancing Delivery Efficiency & Meeting Customer Demands

The respondents raised concerns regarding the increasing demand for faster deliveries and the challenges of meeting these demands with the existing transport offerings in the market, a sentiment supported by Hashimy and Rosines (2021). As a consequence, trucks are often not loaded to full capacity. The significant cost associated with drivers incentives maximising cargo per vehicle, as each pallet's cost per shipment is higher when fewer pallets can offset the driver's expenses. Consequently, many shippers would benefit from utilising smaller trucks. Autonomous trucks offer an opportunity for economic feasibility with smaller vehicles by eliminating driver costs and enabling more frequent pick-ups, ultimately resulting in reduced delivery times.

To further address reducing transit times, shippers, and research (Engström et al. (2019)) commonly regard long-haul transportation as the most advantageous application for ABETs. However, it is worth noting that the ideal solution lies in short-distance, high-frequency transportation — a strategic sweet spot in the transport landscape. These operations are typically conducted on a larger scale by established carriers. Consequently, there are two potential paths for ABET integration: existing carriers adopting autonomous technology to enhance their efficiency or new competitors completely replacing traditional operations with autonomous systems. Given the highly heterogeneous and complex nature of the industry, breaking into the business as a small player is challenging. Additionally, it is unlikely that established carriers would allow a new entrant to handle only a small portion of their operations, as it would raise overall costs. Although respondents recognise the value of long-haul transportation, they acknowledge that the deployment of driverless trucks on highways is not expected to happen in the immediate future.

6.4 Implementation challenges

Customer expectations of autonomous technology may not align with its actual state of development, as noted by Abrahamsson et al. (2023). Media portrayals and exaggerated claims by OEMs contribute to this disconnect. This hinders the realisation of benefits and complicates planning and investment

in automation, as companies lack confidence in technology availability and performance (Hashimy and Rosines, 2021). Furthermore, there is a discrepancy between customers' willingness to pay for innovation and the level of innovation required by the industry. The market is not always ready to pay for a new innovation without clear payback time, especially if the upfront cost is high. Therefore, the shippers are waiting for a large player to make substantial investments to make autonomous solutions efficient and affordable. This was reinforced by the majority of respondents who stated that they are not preparing for automation at all, waiting for it to become widespread before investing. However, there are barriers to investment. It is unclear how an autonomous system would function - whether infrastructure needs to be redesigned with separate lanes or if autonomous vehicles should integrate into the existing road network. Moreover, the requirements for customers at the hub when receiving the truck remain uncertain, adding to hesitations. The current focus in the road freight industry is primarily on electrification, while automation is given less attention. This may be attributed to the perception that everything is working fine, leading to a lack of perceived need for further development.

6.4.1 Challenges of Liability & Public Perception

The issue of liability presents a significant challenge in the adoption of ABETs. Unlike traditional scenarios where a human driver can be held accountable, the use of AI technology and predetermined responses in ABETs complicates the attribution of blame in case of accidents, as highlighted by Fagnant and Kockelman (2015). The interviewees expressed concerns about this uncertainty and stressed the need for clear evidence to establish the trustworthiness of the technology. Overcoming the psychological barrier and changing public perception, as emphasised by Osburg et al. (2022) and Raj et al. (2020), is crucial to ensure that the blame for accidents does not automatically fall on the technology itself, as noted by one of the respondents. Failure to address this challenge could significantly impede the implementation progress and negate the potential benefits of automation. For example, studies such as Fagnant and Kockelman (2015) and Hashimy and Rosines (2021) predict that automation can reduce accidents by eliminating human error. However, the interviewees expressed concerns about the unpredictability of human behaviour and were sceptical that remote operations alone could meet the required safety standards. Consequently, they advocated for piloting autonomous trucks in controlled environments before

full implementation.

6.5 Optimising Implementation: Prioritising Logistics Efficiency & Customer Interaction

When considering the implementation of ABETs, it is crucial to identify areas in the logistics chain where the driver's presence does not provide significant additional value. For this reason, last mile delivery should not be the primary focus of ABET deployment. The driver plays a vital role in customer interaction and satisfaction, which is not adequately captured by existing research. People appreciate being greeted by a person and having the opportunity to communicate directly, enhancing their overall delivery experience. Hence, it is essential to prioritise the integration of ABETs in areas where the driver's customer-facing role can be effectively supplemented without compromising the personalised aspect of the delivery process.

Prioritising the needs of the logistics system will create a transportation network that is both sustainable and profitable. This will require collaboration and innovation from a range of stakeholders, including vehicle manufacturers, electricity companies, logistics companies and policymakers.

Chapter 7

Business Model Analysis: Exploring the Value Proposition of Autonomous Battery Electric Trucks

Based on the analysis, a Value Proposition Canvas was developed to map out the value proposition. In this chapter, the customer profile and value map are presented and described.

7.1 Customer profile

The most important jobs for the shipper are to adapt to the demand for faster deliveries while reducing costs, optimise cargo space, ensure timely and safe delivery, and maintain reliability and accuracy. Shippers also need to comply with regulations, such as driving in low emission zones or handling hazardous goods, while also minimise environmental impact. The customer profile is presented in Figure 7.1.

By minimising delivery time, shippers can meet the growing demand for faster deliveries, offering significant value to their customers. Being flexible and adapting to changing transportation needs is a key aspect of this value proposition. Moreover, gaining increased control over the site is crucial for efficient deliveries. By monitoring the inflow and outflow of goods and knowing the exact timing of deliveries, shippers can adjust their workforce accordingly, resulting in predictable delivery schedules. Additionally,

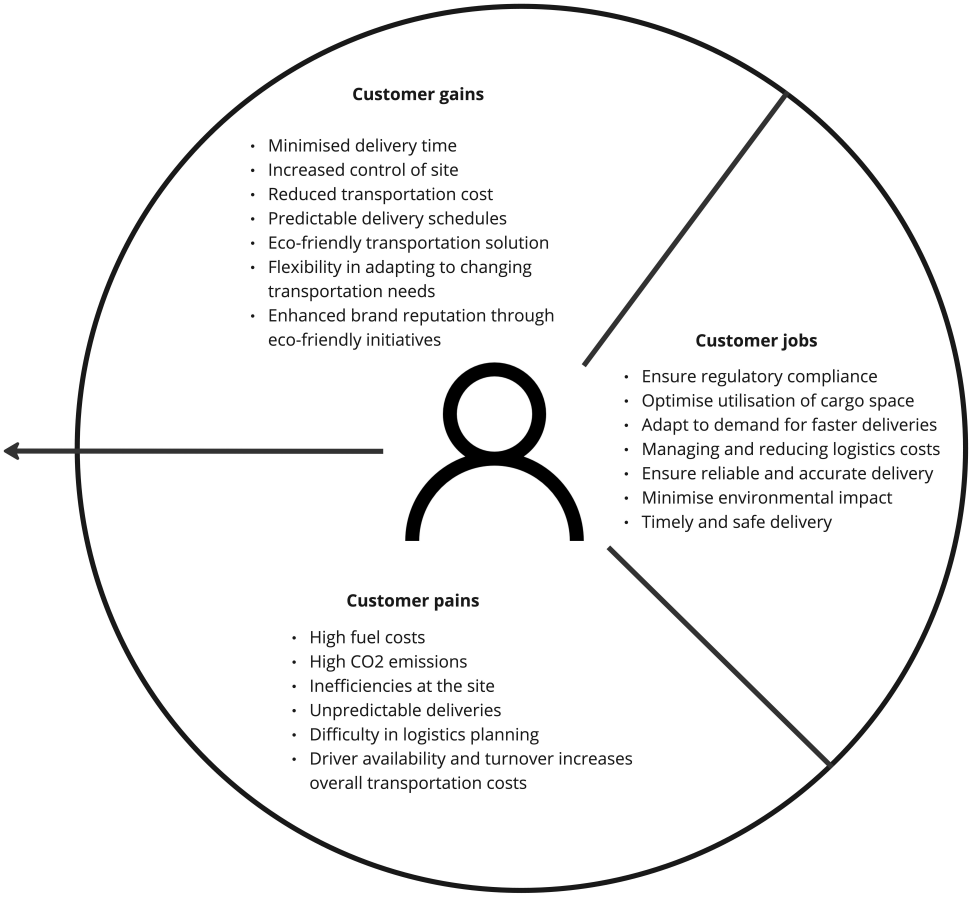


Figure 7.1: Value Proposition Canvas: customer profile

reducing transportation costs is a constant goal for shippers. They aim to maximise the cargo capability of their vehicles, striving to fit more pallets into each truck, which ultimately reduces the cost per pallet. Furthermore, while sustainability may not be their top priority, shippers aspire to be associated with eco-friendly transportation solutions. They understand that enhancing their brand reputation through eco-friendly initiatives can contribute to a positive image in the industry.

High fuel costs pose a significant challenge for those relying on traditional diesel- or gas-powered trucks, causing financial strain and environmental concerns due to high CO2 emissions. Additionally, inefficiencies at the site contribute to further difficulties. Unforeseen deliveries often lead to congestion as shipments arrive in large volumes rather than a steady flow. Furthermore, language barriers frequently result in delays or errors during

operations. Moreover, logistics planning becomes arduous, particularly for e-commerce companies managing an extensive range of products varying in size, shape and weight. The ever-changing market trends and consumer behaviour make it challenging to accurately predict demand and allocate capacity effectively. Furthermore, the shortage of available and reliable drivers exacerbates transportation costs. While shippers may not struggle to find personnel, the increased demand surpasses supply, leading to elevated prices for transportation services. Moreover, drivers incur expenses such as salaries, sick leave, and compliance with new European laws aimed at improving their working conditions. Consequently, transportation companies face rising overhead costs, resulting in increased transportation expenses for shippers.

7.2 Value Map

An autonomous solution provides a unique advantage by offering a logistic service that harnesses advanced technology. Recognising the importance of meeting customer expectations and leveraging the capabilities of existing technology, ABETs ensure precise information and real-time tracking. By implementing nighttime driving, these autonomous systems optimise the flow of deliveries, enabling 24 hours a day, seven days a week delivery capabilities and effectively minimising inefficiencies, addressing congestion issues. This approach results in a flexible and expedited delivery solution that can be tailored to specific routes and schedules. Moreover, the integration of digitisation and supply chain visibility tools enhances control and enables shippers to meet the evolving demands of their customers effectively. The offerings of an autonomous battery electric vehicle can be seen in the value map in Figure 7.2.

Autonomous solutions provide numerous advantages, or gain creators. The integration of autonomy brings about increased digitisation, as autonomous vehicles require connectivity and data transmission. This connectivity enables supply chain visibility, granting shippers the desired level of control. They can access crucial information such as estimated time of arrival and performance measurements, information that can be offered to the customer. Moreover, autonomy offers enhanced flexibility, leading to improved efficiency. By leveraging predictability, routes can be optimised, enabling maximum pickups and expedited deliveries. Additionally, ABETs eliminate the need for breaks,

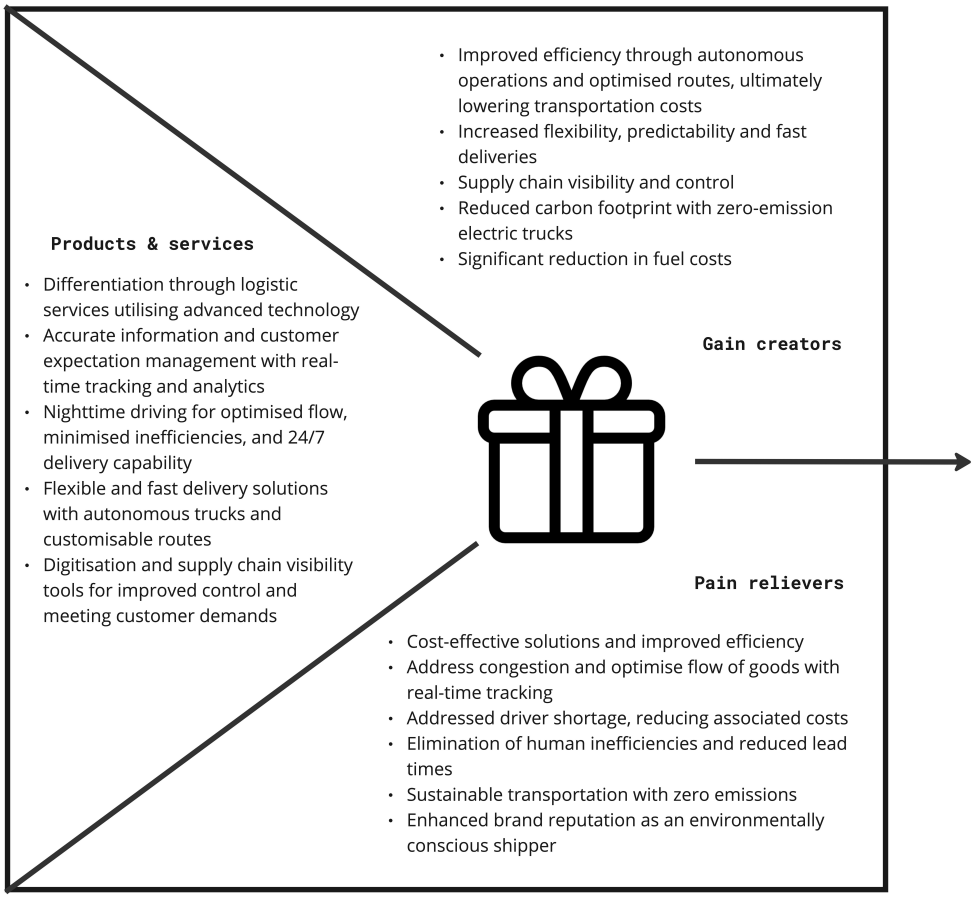


Figure 7.2: Value Proposition Canvas: value map

except for charging time, resulting in faster transportation and ultimately reducing costs. The autonomous solution also contribute to sustainability efforts by significantly reducing fuel consumption and carbon emissions, which combined with electrification creates a significant sustainability advantage.

With an ABET, one of the key pain relievers is the elimination of concerns about driver availability or turnover. This addresses the persistent issue of driver shortage and leads to significant cost reductions associated with hiring and retaining drivers. Moreover, autonomous solutions have the ability to eliminate human inefficiencies, resulting in streamlined operations and reduced transport lead times. This efficiency gain not only lowers operational costs but also enhances overall productivity. Real-time tracking capabilities offered by ABETs further contribute to improved efficiency by reducing

uncertainties in delivery timelines. By adhering to environmental regulations, ABETs provide a sustainable transportation option with zero emissions, effectively addressing the growing desire to reduce reliance on road transport and mitigate carbon footprints. Additionally, as ABETs are electric vehicles, they contribute to a significant reduction in noise pollution, allowing for deliveries in areas with strict noise regulations. This environmental benefit not only demonstrates a commitment to sustainability but also enhances the brand reputation of the shipper as an environmentally conscious entity.

7.3 Navigating Implementation Concerns: Key Considerations for Logistics Service Providers

The value proposition of autonomous solutions is evident, offering several advantages. However, from a LSP perspective, there are still unresolved questions and concerns that need to be addressed.

The widespread adoption of autonomous technology still has a considerable distance to cover, and the industry awaits a major player to embrace and implement it. This delay can be attributed to several factors, including the readiness of technology and regulations to support its integration into long-haul transportation routes, where the potential benefits for shippers are most pronounced. As a result, key questions arise regarding the development, implementation, and target customer of automation. Given that freight companies have the most to gain from enhancing their overall operations, they are well-suited to implement ABETs. Since the majority of shippers rely on existing carriers to handle their shipments, the successful integration of ABETs into the current network becomes crucial for their adoption. This integration would leverage the existing operations and infrastructure, raising the question of an XaaS business model. The study revealed that shippers exhibit hesitancy to invest in a small fleet due to dependence on multiple actors. However, by avoiding vehicle ownership commitments, shippers might be more willing to explore and test autonomous solutions.

To ease the implementation of ABETs, LSPs should allocate resources to effectively define the necessary requirements related to daily operations. For instance, it is essential to consider if the shipper needs to modify their

existing operations to accommodate autonomous vehicles and how the human-machine interaction will be managed. By providing transparent and comprehensive information, LSPs can foster trust and confidence in the technology. Moreover, the design of the vehicle is a crucial aspect that requires careful consideration, as these trucks need to navigate diverse and varying site conditions. For example numerous temporary obstacles on different sites, such as containers and parked cars. Consequently, a one-size-fits-all solution is not feasible, thus the design of the vehicle may require standardised specifications that enables all shippers to adapt their hubs accordingly to enable widespread adoption. However, the study also revealed that hardware components will become readily available for purchase in the future. In light of this, it becomes evident that software capabilities will be the key differentiating factor among various stakeholders and actors operating in the autonomous vehicle market.

7.4 Social Sustainability: Addressing Employment Impact and Working Conditions

Social sustainability remains a crucial aspect to consider in the context of automation. While the removal of a driver position is inevitable with autonomous vehicles, it is important to acknowledge the potential impact on employment. The new role of monitoring the truck may require a higher level of education or advanced skills, which can pose a risk of excluding individuals who are already vulnerable and lack access to the necessary educational resources. Hence, it becomes essential to determine appropriate measures for addressing the unemployment of drivers. The research findings indicate that drivers will still play a role in the value chain, but it is crucial to evaluate and improve their working conditions. Factors such as inflexible work schedules and poor work-life balance contribute to disengagement, isolation and decreased job satisfaction. One potential solution to address these issues is the implementation of remote operations. However, this requires a thoughtful approach that focuses on creating fair and supportive working conditions rather than transferring existing problems to the new role. Instead, proactive measures need to be taken to ensure improvement. The European Union has implemented laws to enhance driver conditions, but these efforts has lead to increased administrative costs for carriers, impacting the economic

sustainability of such solutions. This demonstrates the complexity involved in implementing a radical solution like autonomous vehicles on all aspects of sustainability.

Chapter 8

Conclusions

Shippers display an apparent indifference towards autonomous truck logistics and do not currently prepare for it. However, they view it as an intriguing element of the future of logistics. Despite seeing autonomous logistics as a potential competitive advantage, they believe it primarily serves providers by optimising operations. Fundamentally, basic logistics needs must be met, irrespective of autonomous capabilities. Yet, it is widely acknowledged that logistics service providers (LSP) that fail to adopt autonomous capabilities may struggle to stay competitive against those that do. From a consumer perspective, for instance, the automation of warehouses or trucks by an e-commerce company may not be a primary concern. However, such companies can likely reduce lead times, increase reliability, and maintain cost competitiveness, thereby gaining more market share.

8.1 Assessing the Value of Autonomous Battery Electric Trucks

As automated trucks are increasingly being implemented, shippers anticipate that these technologies will address some of their key challenges. Not all these benefits are directly related to the vehicles themselves, but rather to the necessary digitisation and accompanying developments within the logistics industry. This digital transformation is expected to bring about improved transparency in transportation for shippers, who will gain the ability to live-track their shipments. This will not only enable them to provide accurate estimated arrival times to improve their daily operations, but it will also offer

a value-added service to their end customers by improving the transparency of their logistics.

In addition, shippers expect that automation will reduce inefficiencies at their sites by eliminating the need for human interaction, which can often be hindered by issues such as language barriers or compliance with site procedures. Furthermore, by removing human limitations, a more consistent 24/7 operation becomes possible. This leads to a more balanced traffic flow, potentially reducing congestion at sites and minimising the resources required to handle peak hours of inbound and outbound logistics.

ABETs are also expected to help tackle the rising transport costs, which are driven by increasing fuel prices and driver shortages. They are likely to make operations more predictable in terms of performance and delivery times, as ABETs will operate at a consistent pace. This predictability will further reduce the variable operational costs usually associated with drivers.

Lastly, LSPs that implement ABETs will distinguish themselves in the market by significantly reducing their environmental footprint. While this is often considered a secondary concern for shippers, it is an important factor they consider, largely because it positively impacts their brand reputation among customers and can drive increased revenue. If a LSP can enhance the sustainability of their operations with the help of ABETs, without increasing operational costs, they will gain a significant competitive advantage over competitors who may need to charge a premium or reduce their margins to achieve the same level of sustainability.

8.2 Addressing Frictions & Challenges Related to the Implementation of Autonomous Battery Electric Trucks

Autonomous electric trucks (ABETs) evidently have significant potential to add value within a logistics network. However, there are challenges that need to be addressed effectively to fully realise this potential. One such challenge is not directly related to autonomous driving, but to the digital landscape of the transport industry. For a LSP to offer transport transparency, they need to digitalise the complete transport chain, not just the segment where ABETs are deployed. Even though transport management systems are being adopted

by the industry, these systems need to be digitally integrated with each other. This integration allows a LSP to have a single, trustworthy data source and also enables the ABETs to interact with the next point in the transport chain.

Secondly, there is a noticeable gap between what autonomous logistics can currently offer and what customers, such as shipping companies, expect. This discrepancy, coupled with vaguely defined requirements that these customers must adapt to, breeds hesitation. Primarily, customers are unsure of their responsibilities, the required investment, and the expected return. While it might be more efficient to keep customers minimally involved, LSPs still need to answers to these questions.

Lastly, it seems to be generally agreed that ABETs will bring most value in hub-to-hub operations. However, these typically involve longer distances and public roads, which pose regulatory, technical and safety challenges. These challenges can be particularly tricky and expensive to overcome in the initial deployments of ABETs. It has been suggested that the most feasible initial use case for ABETs would be in high-frequency, short-distance flows, typically close to intermodal hubs or industrial parks.

8.3 Identifying Opportunities and Challenges for a Logistics Service Provider: Recommendations to the Industry

Considering that shippers are primarily concerned with the effective delivery of their goods, rather than whether or not their truck logistics is automated, it is important for a LSP to maintain a focus on meeting these basic needs. However, the benefits of autonomous capabilities should not be overlooked, as early adopters stand to gain significant competitive advantages. To unlock the full potential of transport transparency, initial investments should focus on digital integration across the network, creating interfaces and establishing a single-source-of-truth database.

In addition, a LSP should allocate resources to clearly define the requirements for implementing autonomous trucks. These requirements encompass both physical and digital infrastructure, but it is equally important to consider

operational aspects. LSPs should present these requirements clearly to shippers that may interact with autonomous vehicles, thereby reducing any hesitations.

It is advisable for LSPs to initiate ABET trials within their existing network on high-frequency, short-distance routes where there is a built-in capacity redundancy. Ideally, these routes should require limited involvement from shippers, such as in the case of intermodal hubs. This approach will serve as a test case for ABETs, allowing them to operate in real-world conditions occasionally, and generate insights to define requirements and build a safety case for longer-distance operations.

A recommended approach for LSPs to gain a better understanding of the capabilities and requirements of operating ABETs is to actively participate in EU pilots and research projects. This will give them an opportunity to influence future standards to better align with their solutions, and also to analyse the findings of requirements to further enhance their autonomous offering.

For LSPs with high ambitions for autonomous battery electric vehicles, leveraging the sustainability and innovation aspects of their brand reputation can form a key part of their marketing strategy. This can potentially enable them to secure a larger market share.

In closing, as the logistics industry evolves, it is crucial to understand and prepare for the changes brought by emerging technologies like autonomous logistics. The findings and recommendations in this thesis provide a roadmap for navigating these changes, encouraging the industry to proactively adapt and shape the future of logistics. The implementation of autonomous capabilities promises significant advantages for early adopters and, if managed appropriately, can revolutionise the logistics industry.

Chapter 9

Discussion

This chapter provides a critical analysis of the study, including method, results and analysis. The implications of the findings for theory, practice and future research are discussed, as well as the limitations of the study.

The objective of this study was to comprehensively analyse and assess various aspects of autonomous battery electric trucks in order to gain insights into customer needs, pain points and expectations. By mapping these factors, the aim was to determine the true value proposition of an ABET. Additionally, this research sought to identify and understand the frictions and challenges associated with implementing such a solution in the current market. This study sought to shed light on the opportunities and obstacles that a logistics service provider may face while operating an ABET fleet. By addressing these crucial areas, valuable knowledge and recommendations were generated to propel the advancement of this innovative transportation solution.

9.1 Research Design

The research design employed in this study was specifically used to address the novelty of the research topic, necessitating a significant emphasis on comprehending the subject matter. By dedicating a pre-study to defining the problem, a thorough understanding of the scope was established prior to embarking on the project's data collection phase. This deliberate approach ensured a clear foundation and paved the way for a focused and informed progression throughout the study.

9.1.1 Limitations & Advantages of a Pre-Study Approach

The pre-study aimed to understand the transportation market from a logistics service providers perspective. However, the study was limited as the industry representatives came from the same segment. The shared terminology and common understanding among participants improved data coherence for comparison and analysis, but this approach risked missing out on contrasting viewpoints that could enhance the research topic's overall comprehension. While this provided valuable insights and deepened understanding of the interviewed company's market views, it may not accurately reflect other industries' realities.

The unstructured approach of the pre-study interviews offered the advantage of topic flexibility, allowing for natural and unrestrained conversations. This resulted in unexpected insights and information that may have been overlooked in a structured interview format. However, the absence of predefined questions posed the risk of potential digressions or inclusion of irrelevant information. Some topics explored, like design preferences and technology-specific aspects, may have been redundant or tangential to the research objectives.

Non-participatory observational studies were used to gain insights into natural phenomena like social interactions, behaviours and cultural aspects. To mitigate the Hawthorne effect, where individuals alter their behaviour as a result of being aware of being observed, it was clarified that the purpose was to learn and understand daily operations rather than evaluate the individuals. However, it is important to note that taking pictures during observation may unintentionally influence the behaviour of the observed individuals.

9.1.2 Data collection: Balancing Engagement & Consistency

During the course of the semi-structured interviews, certain participants exhibited a tendency to provide detailed and insightful responses, while others offered only brief answers. This variability in participant engagement have potentially impacted the research findings, as there is a greater likelihood of subjective interpretations or assumptions being made when filling in the

gaps. It was evident that some individuals lacked the requisite knowledge to provide comprehensive insights into specific topics, contributing to the disparity in response lengths. However, the interviews maintained a consistent structure, ensuring that the same set of topics were explored with each participant. This facilitated the generation of comparable data across interviews, enabling robust analysis and interpretation of the findings.

9.1.3 Assessing Bias in Thematic Analysis

Utilising discussions as an analysis method allowed for the exploration of diverse perspectives regarding the interpretations of interviews and existing research. However, it is important to acknowledge that this approach carries a potential risk of introducing subjectivity and bias, as conclusions are derived from individual interpretations. To mitigate this risk, a thematic analysis was used to compare the results and the existing literature, ensuring that the discussions were grounded in the literature findings rather than relying solely on prior knowledge.

9.2 Challenges and Insights in Identifying Specific Customer Needs

Due to the diverse and specific nature of the customer needs that were identified in this study, compiling an in depth, comprehensive list was challenging. Simply listing customers' requests would have resulted in a disorganised collection of words taken out of context. As a result, the analysis and industry recommendations present more general findings that can be applied across companies, such as the desire to lower costs and improve efficiency. It was crucial for this study to gain a deeper understanding of what "lowering costs" entails as most research present seemingly generic results. The findings indicated that each need is interconnected and that lowering costs refers to for example optimising routes and utilising full load capacity. However, attempting to precisely define each factor and ascertain everyone's exact desires becomes an extensive and time-consuming task. This may explain why prior research in this area also tends to be generic. This resulted in some specific aspects, such as lowering CO2 emissions, and some non-specific factors, like flexibility. Thus, it would be recommended

to conduct a case study of a few companies to gain a more profound understanding, rather than attempting to map the entire industry.

Interestingly, the findings closely aligned with existing research. Although the results did not unveil any groundbreaking or unforeseen insights, it was fascinating to observe the remarkable alignment among various sectors within the industry. Even companies operating in distinct domains such as food, construction and hazardous goods exhibited strikingly similar requirements during the interviews.

9.2.1 Validation & Reliability

As autonomous trucks are not widely used, participants expressed their expectations on the technology rather than based on personal experiences, in other words the responses were speculative in nature. Hence, it is not surprising that the findings of the existing research align with the interview responses, considering that the respondents' answers are likely influenced by for example media portrayals and newspaper articles. Consequently, the findings from this study may not accurately represent the reality of autonomous transportation. This emphasises the value of carefully planning studies that provide validity and reliability. The selection of interviewees was conducted randomly and represented various industries, providing a diverse range of knowledge and perspectives. Additionally, the inclusion of industry researchers and external experts allowed for a comprehensive understanding of the automation topic from a wide array of viewpoints. To ensure reliability, this study employed standardised methods consistently across all phases, including data collection, analysis and interpretation. Additionally, the consistency between the literature and the interview data strengthens the validity and credibility of the conclusions drawn, as they are grounded in both academic research and real-world experiences shared by the interviewees.

9.3 Research Contribution & Implications

This study bridges an important gap in the literature by contributing to the current understanding of automation from a customer point of view. This offers valuable guidance for businesses aiming to improve customer satisfaction, adapt their strategies and design user-centric automated systems.

One crucial aspect to consider is the involvement of customers in the development of ABETs. Neglecting their input may lead to a solution that does not align with market demands, rendering it unlikely to be accepted. Therefore, while prioritising technology development is important, it is equally essential to address regulatory, technical and safety concerns to enhance public acceptance and ensure successful market adoption. It is important to note that autonomy is often seen as a radical innovation; however, its implementation should be approached incrementally. By gradually introducing autonomous capabilities, the industry can build trust, mitigate risks, and provide a smoother transition for both customers and stakeholders.

Chapter 10

Limitations & Future Work

Based on this study, it would be beneficial to further develop a comprehensive business model and delve deeper into the LSP perspective. One approach to achieve this could be conducting a case study, which would provide valuable insights. Additionally, it is important to validate the findings of this research with a quantitative study, ensuring the accuracy and reliability of the results. For example, it was challenging to reach a sufficient number of individuals willing to participate in the interviews, which limited the selection of interviewees and resulted in a randomised sample. Therefore, conducting further research with a sample from the same industry could provide a more comprehensive understanding of the topic. Additionally, this constraint hindered the ability to gather multiple perspectives from individuals holding different titles within the same company. Such diverse perspectives could offer valuable insights into customer needs from the acquirer, user and operator standpoints, enabling a better understanding of customer expectations and how they interact with the solution.

10.1 Technical Specifications of Autonomous Vehicles

The primary objective of this study was not to delve into the autonomous technology specifics, thus full autonomy was assumed in the model scenario that was discussed during the interviews. It is important to acknowledge that this approach may have resulted in respondents forming unrealistic expectations, as the envisioned model scenario may differ significantly from

the actual vehicles developed in the future. Nonetheless, this approach provides valuable guidance regarding the customers' expectations for the technology's near-future advancement. However, a future study should focus on exploring the technical specifications of ABETs, delving into factors such as user interface, user experience and human-machine interaction.

10.2 The Significance of Electrification in the Context of Automation

While the primary focus of this study did not center explicitly on electrification, its significance emerged prominently during both the literature review and interviews. This highlights the importance of electrification as a key component of automation. Numerous companies demonstrate a notable emphasis on electrification, likely driven by its relatively comprehensible nature. Consequently, this highlights the need to explore electrification in conjunction with automation. For example, conducting a comparative analysis between automated electric vehicles and non-electric automated vehicles could shed light on the significance of the relationship between electrification and automation.

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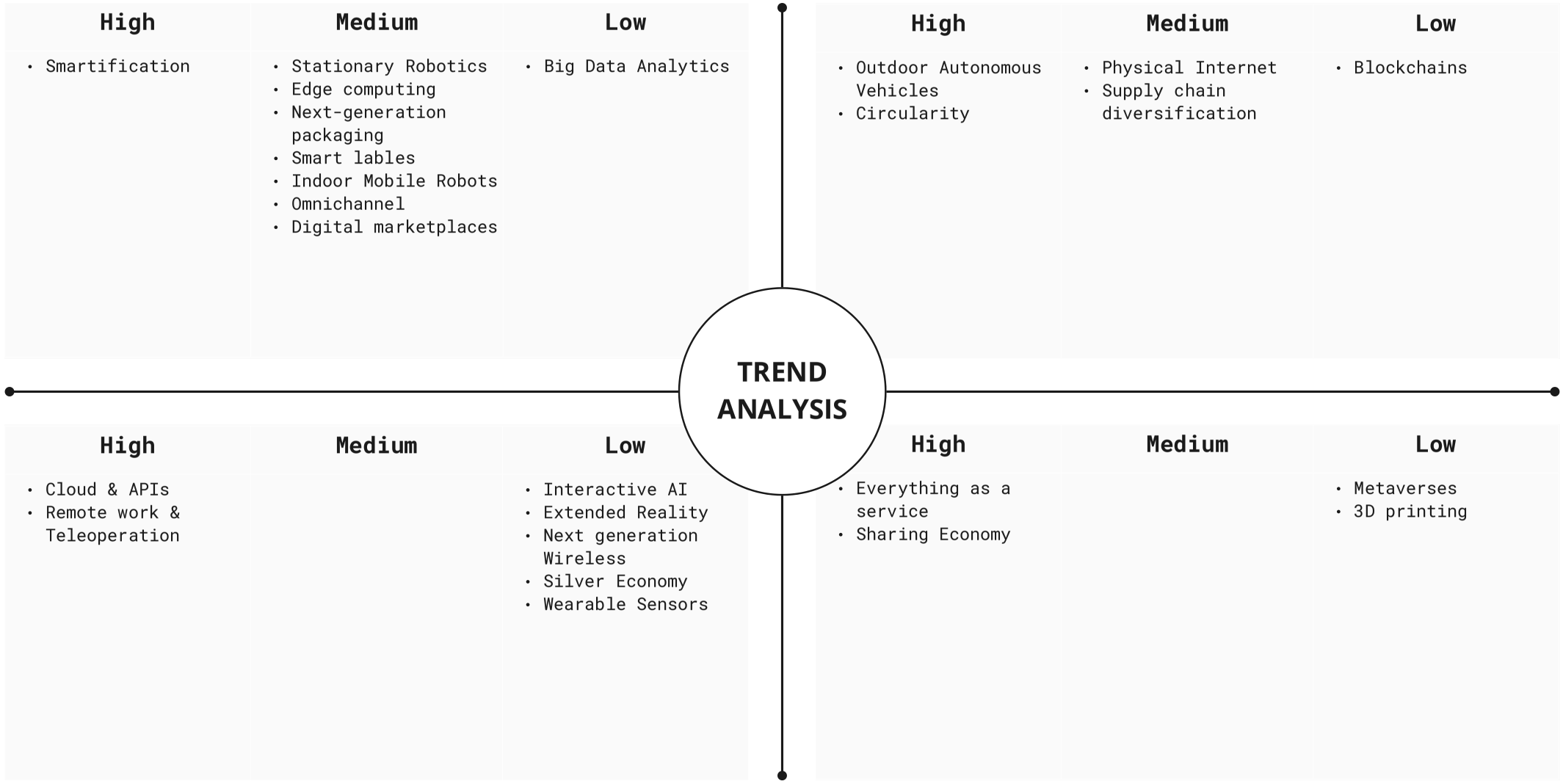
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Appendix A

Trend Analysis

To determine the future of the industry, a trend analysis was conducted using the DHL Logistics Trend Radar (DHL, 2023). The objective was to assess the likelihood of these trends impacting the industry and their connection to automation. Each trend was thus assessed, classified as having low, medium or high relevance. As a result, keywords were identified and further investigated in the literature study.


HIGH IMPACT
LOW IMPACT
TREND ANALYSIS
High
Medium
Low
• Smartification
• Stationary Robotics
• Edge computing
• Next-generation packaging
• Smart lables
• Indoor Mobile Robots
• Omnichannel
• Digital marketplaces
• Big Data Analytics
High
Medium
Low
• Outdoor Autonomous Vehicles
• Circularity
• Physical Internet
• Supply chain diversification
• Blockchains
High
Medium
Low
• Cloud & APIs
• Remote work & Teleoperation
• Interactive AI
• Extended Reality
• Next generation Wireless
• Silver Economy
• Wearable Sensors
High
Medium
Low
• Everything as a service
• Sharing Economy
• Metaverses
• 3D printing
REALISATION 0-5 YEARS
REALISATION 5-10 YEARS

Appendix B

Interview guide

Semi-Structured Interview Guide

Autonomous Electric Road Freight systems
Customer/user needs

Introduction

Interviewee:
Date:
Company:
Title:
Stakeholder level:
Format: *[Online/Offline]*

Background:

We are currently working on our Master's thesis in Innovation Management & Product Development at KTH Royal Institute of Technology to conduct research on Autonomous Electric Road Freight Systems.

The aim of our research is to identify and address the challenges and obstacles that need to be overcome for the successful implementation of autonomous trucks in logistics. The purpose is to help in the development of more efficient and effective autonomous electric freight solutions that meet the needs and expectations of customers.

We would appreciate your input to help us better understand the requirements and expectations of shippers in the road logistics industry. Your feedback will be essential in improving the design and implementation strategy.

Important information:

Please note that this interview is being conducted as part of a Master Thesis project, and everything discussed during the interview may be used within the thesis. However, we would like to assure you that the names of all participants and organizations will be anonymized to protect their privacy and confidentiality.

Is it okay if we record this interview for later reference, since we are writing a report and would like to use the information discussed here as a source? The recording will be used for this purpose only and will be deleted at the latest end of June, when the project has ended.

Question structure

- *Open ended question*
  - *Follow-up questions*
    - *Specific points of interest*

During the course of this semi-structured interview, we may ask follow-up questions and inquire about specific points of interest. These questions are meant to provide support and clarification. Additionally, we will ask "why" questions to gain a deeper understanding of the interviewee's responses and explore related topics.

Introduction **1**
Background: 1
Important information: 1
Question structure 1
1. Interviewee Background & Pain Points **3**
2. Needs **4**
3. Expectations of ABET (Speculative) **5**
4. Model Scenario **6**
Digital Booking 6
Loading/unloading/Goods 6
Palletised goods 7
Container goods 7
Infrastructure/changes on site 7
Daily operations 7
Social aspects 8
Outro **9**

1. Interviewee Background & Pain Points

To start with, we would like to get to know you and the company you work with better. This will make it easier for us to perform a better analysis among our interviewees later.

- Could you tell us a bit about you and your role in the company?
    - How long have you worked at the company and the position you are in right now?
    - What are your responsibilities?
        - Purchasing?
        - Development?
        - Transport Planning?
    - What geographical areas are you working with?
        - Country / region / district / site ?
- Can you describe your daily truck operations in regards to internal flows?
        - How are you solving your shipping demand today? Inhouse / Outsourced?
- Do you find anything challenging?
        - Inefficiencies
    - Is there anything you think is working well?

2. Needs

We would like to better understand the needs and what is important for you, as a shipper, when acquiring & performing your road logistics.

- When considering the purchase of new **transport capacity**, what factors do you take into account?
  - How do you prioritize these factors to make a decision for your organization?
  - Who decides?
  - Who implements?
  - Who are the important stakeholders?
  - How are the different views that stakeholders can have?

3. Expectations of ABET *(Speculative)*

Next, we would like to dive into autonomous electric transports and how you think it should work.

- Do you see any value from adopting autonomous transportation in your daily operations? Why (not)?

- What do you think would be the biggest challenges that need to be addressed before autonomous trucks can be implemented successfully in your internal flows? Why?

- Are you currently working with changes to prepare for autonomous transports? [YES] → How?
  - What is it that usually drives change within the organization?

  - Whose responsibility is it to decide whether to adopt new solutions or modify daily operations?

  - What is usually the primary cause of not implementing a change?
    - resistance to change

- What are your thoughts and opinions on autonomous electric trucks?

4. Model Scenario

From our pre-study, consisting of literature review and expert interviews, we have learned that implementing an Autonomous Electric Road Freight System will most likely require a change in daily operations. We have gathered some examples and would like you to describe how you think this would affect you, challenges related to it and how you think it could be overcome.

The truck is currently driving within and between fenced areas with high automation. This means that there is no human driver in the vehicle, and the truck's sensors and systems are fully capable of handling all driving functions for specific routes and under certain conditions. However, there will still be a remote control driver monitoring the trucks.

Digital Booking

- ❖ When transporting with a driverless and autonomous truck, it would be required to book, schedule and manage the transportation digitally. How would this affect your daily operations?
    - Current digital level
        - Transport planning
        - Shipment tracking
        - Communication
    - Current autonomous level
    - Future digitalisation

    - ➢ What type of systems would need to be integrated?
        - Planning tools
        - Goods management

    - ➢ What would happen if you remove the physical paper?
        - Integration in the logistics chain

Loading/unloading/Goods

- ❖ When the vehicle arrives at the loading area and is ready to be loaded/unloaded, there will be no truck driver on site to manage this. What are your initial thoughts on this?

    - ➢ How do you think that the interaction between the ground personnel and the autonomous truck would look like at the loading bay/area?
        - Errors in loading
        - Changes

Palletised goods	Container goods
The first generations of ABETs will most likely be **rigid** trucks with a **smaller load capacity**. ❖ Could you tell us a bit what this would mean for your operations? ➢ What are the challenges? ➢ What could be the gains?	The first generations of ABETs would most likely be **flatbed** trucks with a **smaller load capacity**. ❖ Could you tell us a bit what this would mean for your operations? ➢ What are the challenges? ■ How do you think the challenges could be addressed? ➢ What could be the gains?

Infrastructure/changes on site

One way to ensure safe and reliable implementation of autonomous vehicles in the near future is to reduce complexity in the operating environment.

- ❖ What types of changes to the site are possible to make?
  - ➢ How do you see that people and other traffic can be avoided?
    - Markings for truck
    - Separate lanes
    - Site regulations

Daily operations

Maintenance involves conducting daily check-ups to ensure everything is functioning as intended. If an issue is detected during the day, a technician is notified to come and check the truck. In the event of a breakdown, another truck will take over operations, minimizing downtime and making sure that deliveries are made on schedule.

- ❖ What are your thoughts on this?
  - ➢ Do you see any flaws?

  - ➢ What are your expectations regarding the steps that will be taken in the event of a truck breakdown?
    - Response time
    - Shifting in between from Autonomous to regular truck

Social aspects

- Replacing your fleet with autonomous vehicles would take place in steps and thus parts of your fleet would still be operated like it is today. How do you think this would affect the partnerships with your current carriers?
    - In what ways do you think your existing carriers might react or adapt to the incorporation of autonomous vehicles into your fleet?
    - How can a logistics service provider work to ensure that existing relationships and solutions are not jeopardized?
- What changes would you be willing to make based on the value you think you will get out of an autonomous solution?
    - Which changes are easy/hard to make?
    - Who is the decision reliant on?

Outro

- Is there anything else that you would like to share or any questions that you were hoping we would ask, but we haven't covered yet?
  - Would it be okay for us to reach out to you after the interview in case we have additional questions?

- Do you happen to know of anyone else who has relevant expertise or experience in this field that we could reach out to for additional insights or perspectives?

Thank you so much for taking the time to speak with us today. Your insights have been truly valuable and we are grateful for the opportunity to learn from you!

If you have any further questions or if something comes to mind after this, please don't hesitate to reach out.

Thanks again for your time and have a great day!

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